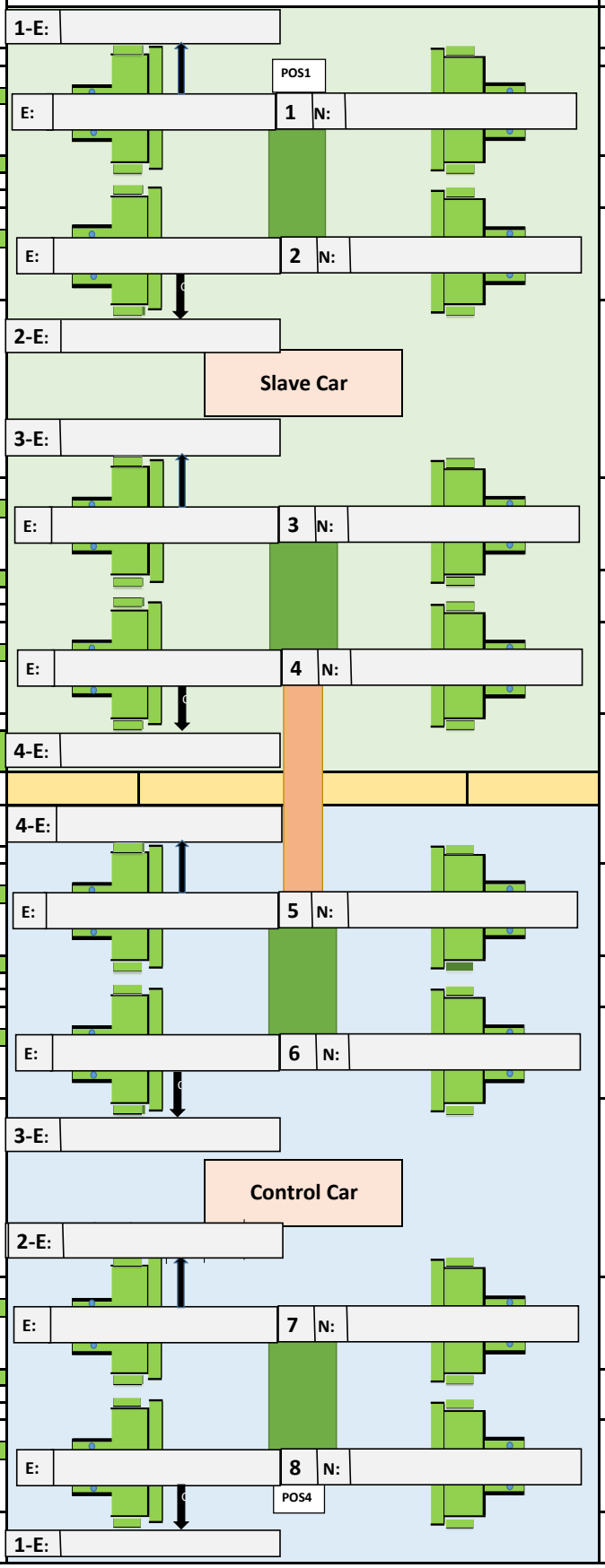
Date:		B20020		PMO2 Work	
North	Road	South		Wagon Recall Road	86
				Bearing- Installed	SG / K Class

	Wheel	Bearing
L1	No Defect	No Defect
Bogie 1	No Defect	
L2	No Defect	No Defect
L3	No Defect	No Defect
Bogie 2	No Defect	
L4	No Defect	No Defect
Wagon Pair Brake Fault	No Defect	
L5	No Defect	No Defect
Bogie 3	No Defect	
L6	No Defect	No Defect
L7	No Defect	No Defect
Bogie 4	No Defect	
L8	No Defect	No Defect

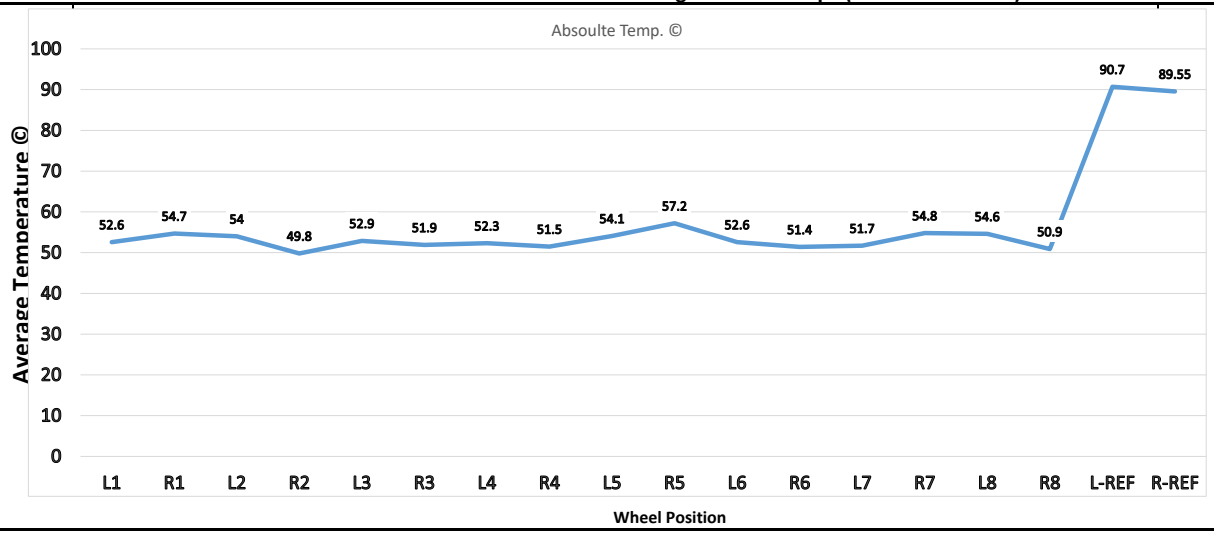


	Wheel	Bearing
R1	No Defect	No Defect
R2	No Defect	No Defect
R3	No Defect	No Defect
R4	No Defect	No Defect
R5	No Defect	No Defect
R6	No Defect	No Defect
R7	No Defect	No Defect
R8	No Defect	No Defect

Note : PTO for Axle Level Brake Performance Ilustration -Average of Last N Trips (Brake Test Series).

B20020

Axle Level Brake Performance Illustration- Average of Last N Trips (Brake Test Series)



HBD -AAR Data								
Wagon No.:	B20020							
	HBD WL		HBD Bearing		RIM AAR		Flange AAR	
Position	L -Side	R-Side	L -Side	R-Side	L -Side	R-Side	L -Side	R-Side
1	167	183	98	115				
2	173	230	98	115				
3	187	179	98	114				
4	178	169	98	115				
5	210	201	98	115				
6	178	191	98	115				
7	218	202	98	115				
8	195	168	98	115				

List of Notifications attached to the Workpack

20020					
Sr.No.	Order No	Notificatior	Notification Description	Completed Work	Comment- If any
1	74396212	72187381	very minor air leak from ECP manifold	Yes/No	
2	74396212	72412119	Brake System Failed	Yes/No	
3	115414358	72177766	Wagon Combined Reservoir STAT Insp	Yes/No	
4	74377116	72358547	Install label on the 98L Combined Tanks	Yes/No	
5	115407468	72170093	Wagon Supply Reservoir STAT Insp	Yes/No	
6	74384529	72365912	Install Label on the 165L Supply Tank	Yes/No	
7	73188436	67648315	R5 Tread-Recall RD88	Yes/No	
8	73188436	67648404	R3 Tread-Recall RD88	Yes/No	
9	73188436	67648532	L5 Tread-Must cut RD88	Yes/No	

Rio Tinto

MAINTENANCE ORDER

115407468

PICS Railway Seven Mile

STATUTORY

Order Description	Wagon Supply Reservoir STAT Insp	
FunctLocation	3076WAG20020 .ABRK.ARSR	Air Receiver: Supply Reservoir
Productive Unit	3076WAG20020	Ore Wagon Pair B Type
Location	2RI0212030	Rolling stock maintenance workshop
Equipment		
Assembly		
Order Type	PM01	Primary Maintenance
Notification No.	**72170093**	Wagon Supply Reservoir STAT Insp
Sort Field		
Planner Group	RW1	Wagons

Order Start Date	20.04.2026	**Order End Date**	20.04.2026
Priority	A	High Priority	
ABC Indicator (Criticality)	3	Class III - High	
Revisión	W7WG2617	RAIL 7 Mile Wagons 2026 Week 17	
Task List	A 319149 1	**Plan**	2094577
TL Critical Risk Assessed			

Order Long Text

Wagon Supply Reservoir STAT Insp

Supply reservoirs + combined reservoirs

To adhere to AS3711 Table 4.1 (2/4Y Inspection) however adjusted to 6Y & 12Y based on Wood RBI 199769-IM-REP-001

OpNo	**WorkCtr**	**Task Description**	**Duration**	**Actual**	
Functional Location			**No. People**		
Notification	**System Cond.**	**Operation Scheduled Start**	**Statutory**	**PSR**	**FSR**
Team Name		**Generic Field**			
0010	RMW	6Y Wagon Supply Reservoir Ext STAT Insp	0,0 H	_____	
3076WAG20020	.ABRK.ARSR	Air Receiver: Supply Reservoir	0	_____	
72170093	On or Off Line	20.04.2026 00:00:00	X		
	Document	6Yr Wagon SR PV Ext Stat Insp 10000392108 ZMP 000 00			
0020	RMW	12Y Wagon Supply Reservoir Int STAT Insp	0,0 H	_____	
			0	_____	
	Off line	20.04.2026 00:00:00	X		
	Document	12Yr Wagon SR PV Int Stat Insp 10000392110 ZMP 000 00			

Rio Tinto

MAINTENANCE ORDER

115407468

PICS Railway Seven Mile

STATUTORY

Order Description	Wagon Supply Reservoir STAT Insp	
FunctLocation	3076WAG20020 .ABRK.ARSR	Air Receiver: Supply Reservoir
Productive Unit	3076WAG20020	Ore Wagon Pair B Type
Location	2RI0212030	Rolling stock maintenance workshop
Equipment		
Assembly		
Order Type	PM01	Primary Maintenance
Notification No.	72170093	Wagon Supply Reservoir STAT Insp
Sort Field		
Planner Group	RW1	Wagons

Information below is to be SAP Text

Comments

Name _____ Signature _____

Supervisor's Name _____ Supervisor's Signature _____

Downtime details:

Date Started:__/__/__ Time Started:__:__ Date Finished:__/__/__ Time Finished:__:__

RTIO-LO-1055376

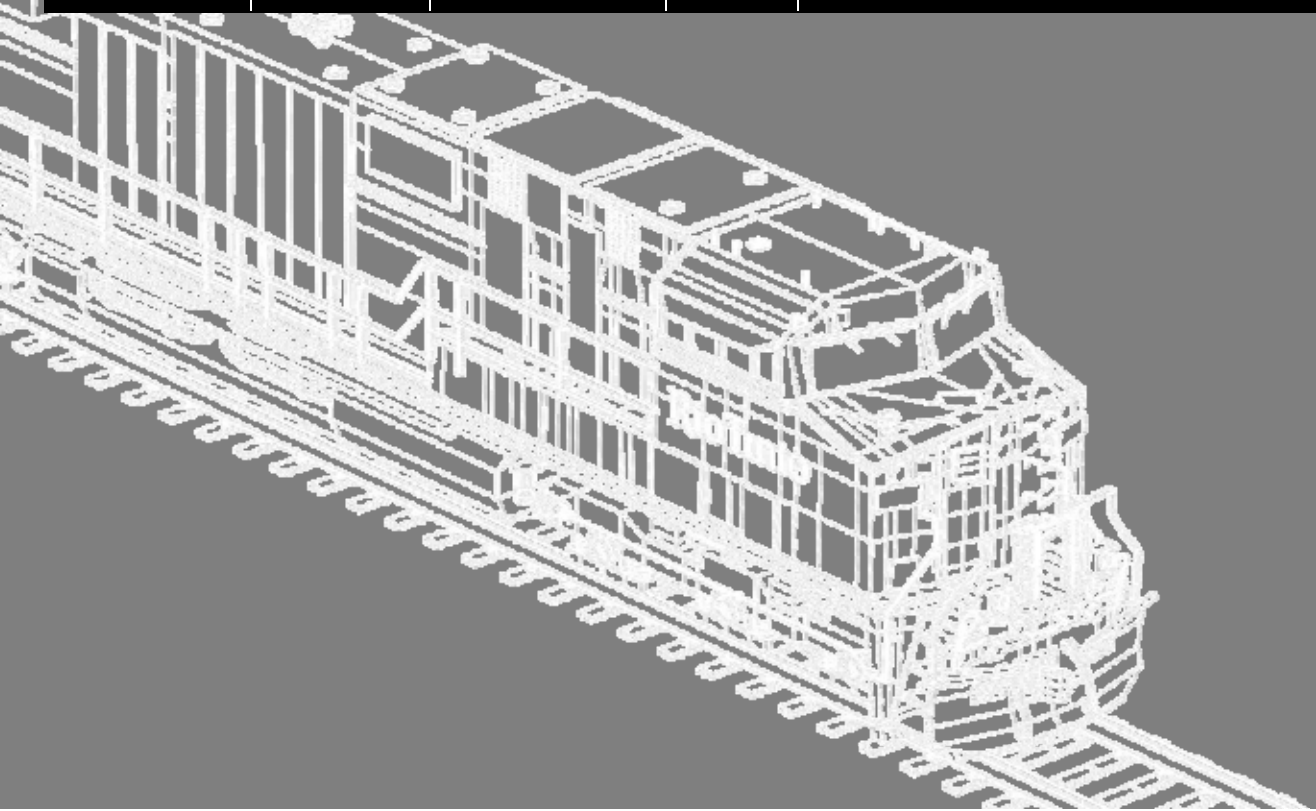
6Yr Wagon Supply Reservoir (SR) PV External Stat Insp

Amendment record

This table identifies the changes to the document made for each new version.

Document Control and revision history RTIO-AM-1047070				Document type: Management Procedure	
Version	**Date**	**Author**	**Reviewer**	**Approved**	**Approved for use from**
1.0	21 June 2024	Soham Patel	Harsh Patel	Brad McCarthy	

Amendment Record				Document Area of Accountability: GM Railways Division
Version & Date	**Author**	**Change MOC #**	**Section #**	**Modification Description**
				•



Purpose and Scope

AIM The aim of this task is to meet the statutory requirements in accordance with the Risk Based Inspection regime referenced in the below SWP.

INSTRUCTIONS Mechanical maintainer to complete tests as per the attached document.

ISOLATIONS: Isolations as per Ironsafe standards.
Refer to SWP's for relevant Hazards and controls
Follow PRC-RP-I-178 Area Isolation Procedure 7 Mile and 8 Mile RSM Wagons
Follow PRC-RP-I-023 Area Isolations Procedures (Cape Lambert)
Conduct Take 5 and Critical Control Checklist.

PPE REQUIREMENTS: PPE as per Ironsafe standards.

TOOLS & EQUIPMENT: Tools and equipment to comply with Ironsafe standards.

SPECIAL NOTES All maintainers shall tick all operations that they complete on the service sheet and sign off. Completed sheets are to be handed to the relevant Supervisor..

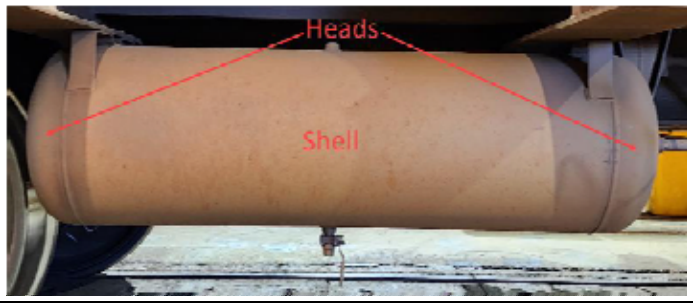
Reference Documents	FDMS No:
1. SWP-6Yr Wagon SR & CR PV External Stat Insp	RTIO-1041970

Documentation Sign Off	Notification Follow Up Required	YES/NO
	Document Update Required	YES/NO
Name:	Sign:	Date:
Supervisor:	Signature:	Date:
Comments:		

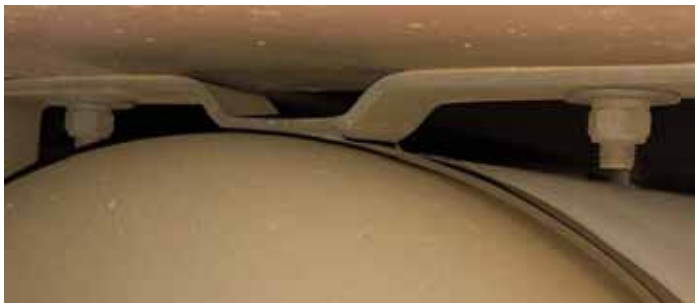
Internal Use Only
@2024 Rio Tinto





Check list	Initial
☐ Record the serial number of Supply Reservoir (SR) pressure vessel. Serial number | 	
☐ Conduct a close visual inspection of the external surface coating/paintwork for degradation such as: ○ General breakdown ○ Flaking or chipping ○ Discolouration or oxidisation ○ Blistering ○ Scars ○ Rusting of substrate and rust spots ○ Record any areas where the coating requires remediation.  	
☐ Conduct a close visual inspection of the shell and heads (dished ends) condition, looking for: ○ Mechanical damage or dents ○ Corrosion ○ Pitting ○ Bulging ○ Wall loss ○ Wear ○ Cracking ○ Inspect welds and area immediately adjacent for cracking or surface pitting. 	

Internal Use Only
@2024 Rio Tinto

☐ Conduct a close visual inspection of the connected pipework for Supply Reservoir (SR) for any signs of: ○ Dents ○ Cracks ○ Gouges ○ Deformation ○ Bulges ○ Leaks, weeps, and seeps ○ Corrosion ○ Inspect welds and area immediately adjacent for cracking or surface pitting. 	
☐ Conduct a close visual inspection of the vessel mounting bracket supports for any signs of: ○ Misalignment ○ Cracks ○ Deformation ○ Gouges ○ Dents ○ Corrosion ○ Fretting ○ Inspect welds and area immediately adjacent for cracking or surface pitting. 	
☐ Conduct a close visual inspection of the bolting and other connections for signs of degradation including: ○ Corrosion ○ Dents ○ Cracking ○ Distortion ○ Staining ○ Mechanical damage ○ Loose or missing bolt 	
☐ If Supply Reservoir (SR) pressure vessel exhibit any signs of damage/degradation, inform the workshop supervisor.	

Comments:

Internal Use Only
@2024 Rio Tinto

RTIO-LO-1055731

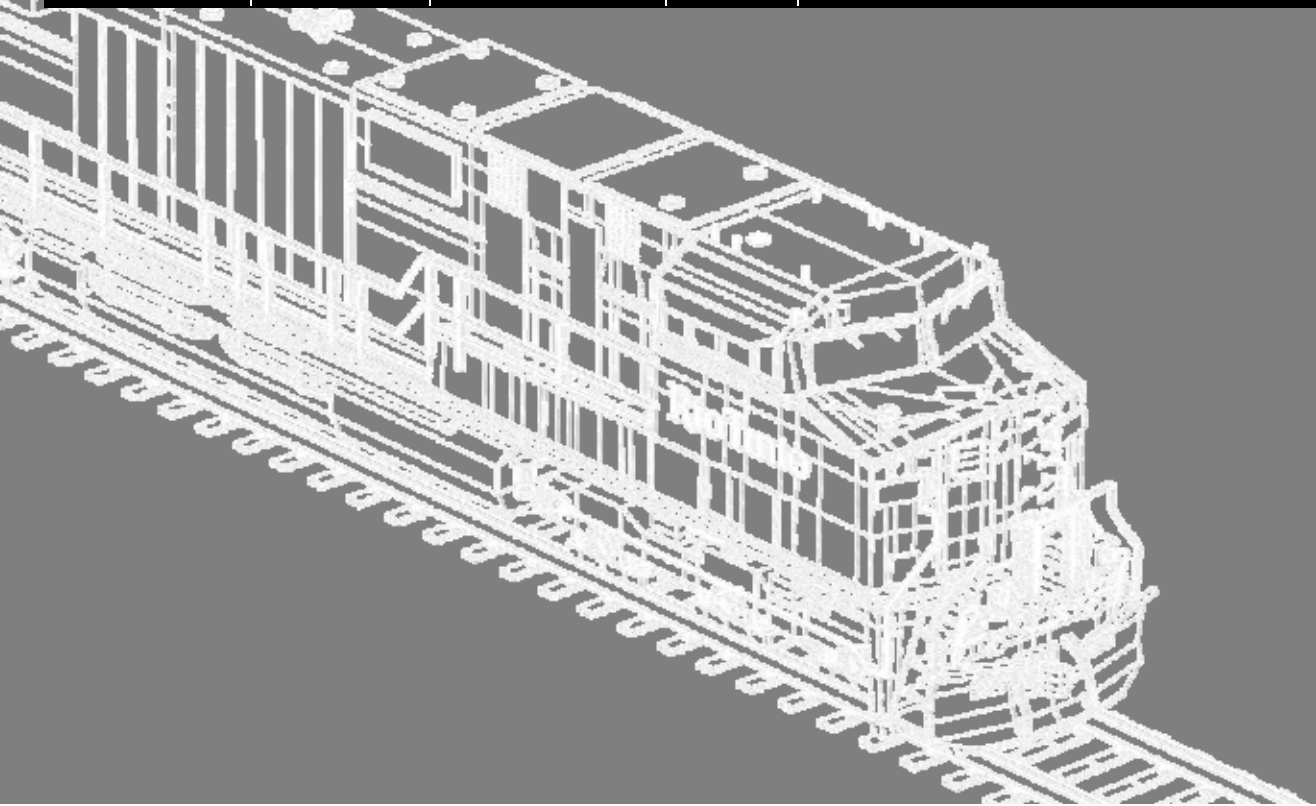
12Yr Wagon Supply Reservoir (SR) PV Internal Stat Insp (Non-Intrusive)

Amendment record

This table identifies the changes to the document made for each new version.

Document Control and revision history RTIO-AM-1047070				Document type: Management Procedure	
Version	**Date**	**Author**	**Reviewer**	**Approved**	**Approved for use from**
1.0	23 June 2024	Soham Patel	Harsh Patel	Brad McCarthy	

Amendment Record				Document Area of Accountability: GM Railways Division
Version & Date	**Author**	**Change MOC #**	**Section #**	**Modification Description**
				•



Purpose and Scope

AIM — The aim of this task is to meet the statutory requirements in accordance with the Risk Based Inspection regime referenced in the below SWP.

INSTRUCTIONS — Mechanical maintainer to complete tests as per the attached document.

ISOLATIONS: — Isolations as per Ironsafe standards.
Refer to SWP's for relevant Hazards and controls
Follow PRC-RP-I-178 Area Isolation Procedure 7 Mile and 8 Mile RSM Wagons
Follow PRC-RP-I-023 Area Isolations Procedures (Cape Lambert)
Conduct Take 5 and Critical Control Checklist.

PPE REQUIREMENTS: — PPE as per Ironsafe standards.

TOOLS & EQUIPMENT: — Tools and equipment to comply with Ironsafe standards.

SPECIAL NOTES — All maintainers shall tick all operations that they complete on the service sheet and sign off. Completed sheets are to be handed to the relevant Supervisor..

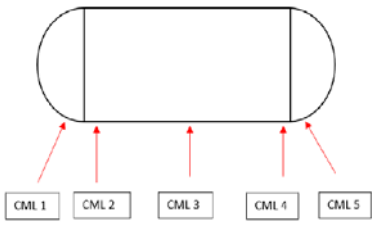
Reference Documents	FDMS No:
1. SWP - 12YR Wagon SR & CR PV Internal Stat Insp (Non-Intrusive)	RTIO-1041975

Documentation Sign Off	Notification Follow Up Required	YES/NO
	Document Update Required	YES/NO
Name:	**Sign:**	**Date:**
Supervisor:	**Signature:**	**Date:**
Comments:		

Internal Use Only
@2024 Rio Tinto





Check list	Initial
☐ Confirm that calibration of ultrasonic thickness tester is done prior to thickness test as per SWP RTIO-1041975.	
☐ Identify location of corrosion measurement locations (CML's) of Supply Reservoir (SR). ○ CML1: Low point of the left dished end straight flange section. ○ CML2: Low point of shell (left side). ○ CML3: Low point of shell (centre). ○ CML4: Low point of shell (right side). ○ CML5: Low point of the right dished end straight flange section. 	
☐ Record CML results. Note: If the minimum remaining wall thickness is less than the anomaly trigger point, confirm this value with a second measurement at the same location. If second measurement confirms an anomaly, proceed to report this to the workshop supervisor.	

Measurement Location	Diameter (mm)	Nominal Wall Thickness (mm)	Anomaly Trigger Point (mm)	Minimum Measurement Found (mm)
SR CML1	406.4	2.39	1.97	
SR CML2	406.4	2.85	2.53	
SR CML3	406.4	2.85	2.53	
SR CML4	406.4	2.85	2.53	
SR CML5	406.4	2.39	1.97	

Internal Use Only
@2024 Rio Tinto

Comments:

Internal Use Only
@2024 Rio Tinto

Rio Tinto

MAINTENANCE ORDER

115414358

PICS Railway Seven Mile

STATUTORY

Order Description	Wagon Combined Reservoir STAT Insp		
FunctLocation	3076WAG20020 .ABRK.ARCR	Air Receiver: Combined Reservoir	
Productive Unit	3076WAG20020	Ore Wagon Pair B Type	
Location	2RI0212030	Rolling stock maintenance workshop	
Equipment			
Assembly			
Order Type	PM01	Primary Maintenance	
Notification No.	**72177766**	Wagon Combined Reservoir STAT Insp	
Sort Field			
Planner Group	RW1	Wagons	

Order Start Date	20.04.2026	**Order End Date**	20.04.2026
Priority	A	High Priority	
ABC Indicator (Criticality)	3	Class III - High	
Revisión	W7WG2617	RAIL 7 Mile Wagons 2026 Week 17	
Task List	A 319149 2	**Plan**	2100903
TL Critical Risk Assessed			

OpNo	**WorkCtr**	**Task Description**	**Duration**	**Actual**	
Functional Location			**No. People**		
Notification	**System Cond.**	**Operation Scheduled Start**	**Statutory**	**PSR**	**FSR**
Team Name		**Generic Field**			
0010	RMW	6Y Wagon Combined Reserv Ext STAT Insp	0,0 H	_____	
3076WAG20020	.ABRK.ARCR	Air Receiver: Combined Reservoir	0	____	
72177766	Off line	20.04.2026 00:00:00	X		
Document		6Yr Wagon CR PV Ext Stat Insp 10000392109 ZMP 000 00			
0020	RMW	12Y Wagon Combined Reserv Int STAT Insp	0,0 H	_____	
			0	____	
	Off line	20.04.2026 00:00:00	X		
Document		12Yr Wagon CR PV Int Stat Insp 10000392111 ZMP 000 00			

Rio Tinto

MAINTENANCE ORDER

115414358

PICS Railway Seven Mile

STATUTORY

Order Description	Wagon Combined Reservoir STAT Insp	
FunctLocation	3076WAG20020 .ABRK.ARCR	Air Receiver: Combined Reservoir
Productive Unit	3076WAG20020	Ore Wagon Pair B Type
Location	2RI0212030	Rolling stock maintenance workshop
Equipment		
Assembly		
Order Type	PM01	Primary Maintenance
Notification No.	72177766	Wagon Combined Reservoir STAT Insp
Sort Field		
Planner Group	RW1	Wagons

Information below is to be SAP Text

Comments

Name _____ Signature _____

Supervisor's Name _____ Supervisor's Signature _____

Downtime details:

Date Started:__/__/__ Time Started:__:__ Date Finished:__/__/__ Time Finished:__:__

RTIO-LO-1055396

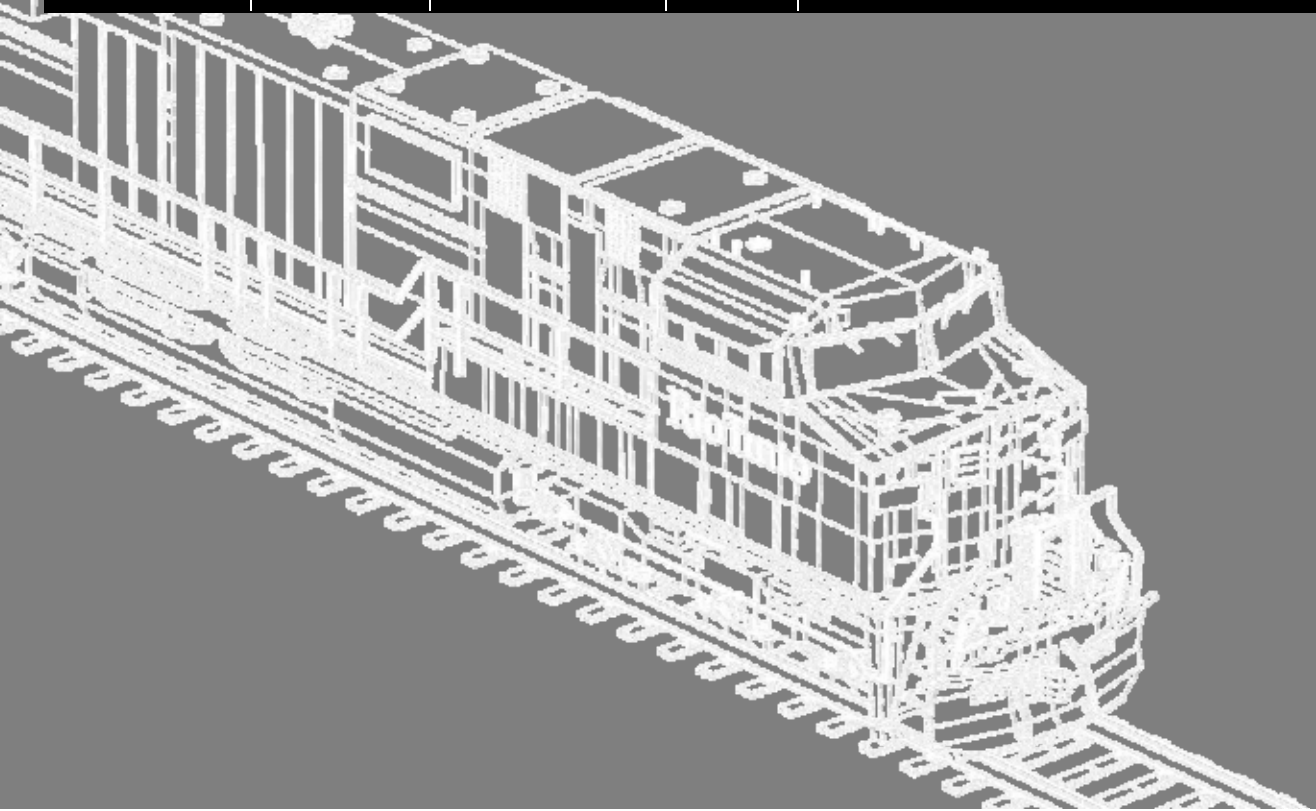
6Yr Wagon Combined Reservoir (CR) PV External Stat Insp

Amendment record

This table identifies the changes to the document made for each new version.

Document Control and revision history RTIO-AM-1047070				Document type: Management Procedure	
Version	**Date**	**Author**	**Reviewer**	**Approved**	**Approved for use from**
1.0	21 June 2024	Soham Patel	Harsh Patel	Brad McCarthy	

Amendment Record				Document Area of Accountability: GM Railways Division
Version & Date	**Author**	**Change MOC #**	**Section #**	**Modification Description**
				•



Purpose and Scope

AIM — The aim of this task is to meet the statutory requirements in accordance with the Risk Based Inspection regime referenced in the below SWP.

INSTRUCTIONS — Mechanical maintainer to complete tests as per the attached document.

ISOLATIONS: — Isolations as per Ironsafe standards.
Refer to SWP's for relevant Hazards and controls
Follow PRC-RP-I-178 Area Isolation Procedure 7 Mile and 8 Mile RSM Wagons
Follow PRC-RP-I-023 Area Isolations Procedures (Cape Lambert)
Conduct Take 5 and Critical Control Checklist.

PPE REQUIREMENTS: — PPE as per Ironsafe standards.

TOOLS & EQUIPMENT: — Tools and equipment to comply with Ironsafe standards.

SPECIAL NOTES — All maintainers shall tick all operations that they complete on the service sheet and sign off. Completed sheets are to be handed to the relevant Supervisor..

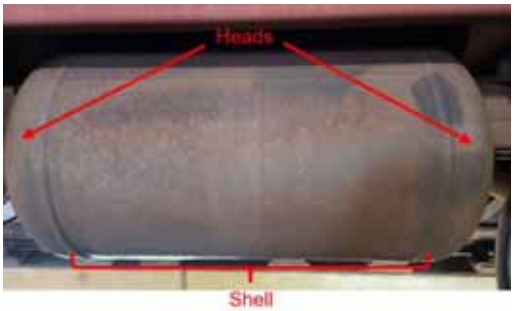
Reference Documents	FDMS No:
1. SWP-6Yr Wagon SR & CR PV External Stat Insp	RTIO-1041970

Documentation Sign Off	Notification Follow Up Required	YES/NO
	Document Update Required	YES/NO
Name:	**Sign:**	**Date:**
Supervisor:	**Signature:**	**Date:**
Comments:		

Internal Use Only
@2024 Rio Tinto





Check list	Initial
☐ Record the serial number of Combined Reservoir (CR) pressure vessel. Serial number | 	
☐ Conduct a close visual inspection of the external surface coating/paintwork for degradation such as: ○ General breakdown ○ Flaking or chipping ○ Discolouration or oxidisation ○ Blistering ○ Scars ○ Rusting of substrate and rust spots ○ Record any areas where the coating requires remediation.  	
☐ Conduct a close visual inspection of the shell and heads (dished ends) condition, looking for: ○ Mechanical damage or dents ○ Corrosion ○ Pitting ○ Bulging ○ Wall loss ○ Wear ○ Cracking ○ Inspect welds and area immediately adjacent for cracking or surface pitting. 	
☐ Conduct a close visual inspection of the connected pipework for Combined Reservoir (CR) for any signs of:	

Internal Use Only
@2024 Rio Tinto

- Dents
- Cracks
- Gouges
- Deformation
- Bulges
- Leaks, weeps, and seeps
- Corrosion
- Inspect welds and area immediately adjacent for cracking or surface pitting.



☐ Conduct a close visual inspection of the vessel mounting bracket supports for any signs of:

- Misalignment
- Cracks
- Deformation
- Gouges
- Dents
- Corrosion
- Fretting
- Inspect welds and area immediately adjacent for cracking or surface pitting.





☐ Conduct a close visual inspection of the bolting and other connections for signs of degradation including:

- Corrosion
- Dents
- Cracking
- Distortion
- Staining
- Mechanical damage
- Loose or missing bolt







☐ If Combined Reservoir (CR) pressure vessel exhibit any signs of damage/degradation, inform the workshop supervisor.

Comments:

Internal Use Only
@2024 Rio Tinto

Internal Use Only
@2024 Rio Tinto

RTIO-LO-1055732

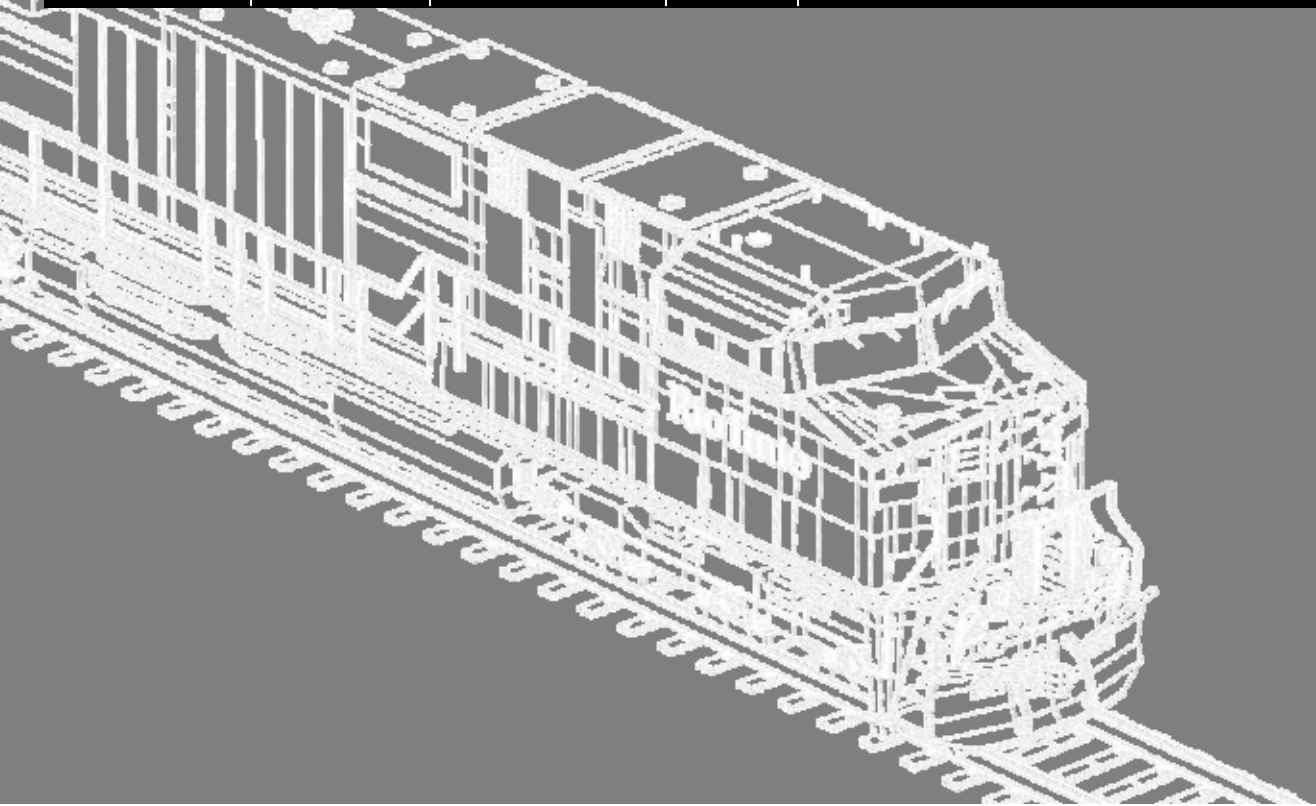
12Yr Wagon Combined Reservoir (CR) PV Internal Stat Insp (Non-Intrusive)

Amendment record

This table identifies the changes to the document made for each new version.

Document Control and revision history RTIO-AM-1047070				Document type: Management Procedure	
Version	**Date**	**Author**	**Reviewer**	**Approved**	**Approved for use from**
1.0	23 June 2024	Soham Patel	Harsh Patel	Brad McCarthy	

Amendment Record				Document Area of Accountability: GM Railways Division
Version & Date	**Author**	**Change MOC #**	**Section #**	**Modification Description**
				•



Purpose and Scope

AIM — The aim of this task is to meet the statutory requirements in accordance with the Risk Based Inspection regime referenced in the below SWP.

INSTRUCTIONS — Mechanical maintainer to complete tests as per the attached document.

ISOLATIONS: — Isolations as per Ironsafe standards.
Refer to SWP's for relevant Hazards and controls
Follow PRC-RP-I-178 Area Isolation Procedure 7 Mile and 8 Mile RSM Wagons
Follow PRC-RP-I-023 Area Isolations Procedures (Cape Lambert)
Conduct Take 5 and Critical Control Checklist.

PPE REQUIREMENTS: — PPE as per Ironsafe standards.

TOOLS & EQUIPMENT: — Tools and equipment to comply with Ironsafe standards.

SPECIAL NOTES — All maintainers shall tick all operations that they complete on the service sheet and sign off. Completed sheets are to be handed to the relevant Supervisor..

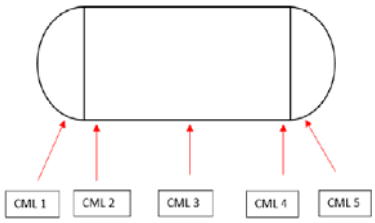
Reference Documents	FDMS No:
1. SWP - 12YR Wagon SR & CR PV Internal Stat Insp (Non-Intrusive)	RTIO-1041975

Documentation Sign Off	Notification Follow Up Required	YES/NO
	Document Update Required	YES/NO
Name:	Sign:	Date:
Supervisor:	Signature:	Date:
Comments:		

Internal Use Only
@2024 Rio Tinto





Check list	Initial
☐ Confirm that calibration of ultrasonic thickness tester is done prior to thickness test as per SWP RTIO-1041975.	
☐ Identify location of corrosion measurement locations (CML's) of Combined Reservoir (CR). ○ CML1: Low point of the left dished end straight flange section. ○ CML2: Low point of shell (left side). ○ CML3: Low point of shell (centre). ○ CML4: Low point of shell (right side). ○ CML5: Low point of the right dished end straight flange section. 	
☐ Record CML results. Note: If the minimum remaining wall thickness is less than the anomaly trigger point, confirm this value with a second measurement at the same location. If second measurement confirms an anomaly, proceed to report this to the workshop supervisor.	

Measurement Location	Diameter (mm)	Nominal Wall Thickness (mm)	Anomaly Trigger Point (mm)	Minimum Measurement Found (mm)
CR CML1	416.05	4.50	3.02	
CR CML2	416.05	3.73	2.97	
CR CML3	416.05	3.73	2.97	
CR CML4	416.05	3.73	2.97	
CR CML5	416.05	4.50	3.02	

Internal Use Only
@2024 Rio Tinto

Comments:

Internal Use Only
@2024 Rio Tinto

MAINTENANCE ORDER

73188436

PICS Railway Seven Mile

Order Description	L5 Tread-Must cut RD88		
FunctLocation	3076WAG20020	.WSET	Wheelset Assembly
Productive Unit	3076WAG20020		Ore Wagon Pair B Type
Location	2RI0212030		Rolling stock maintenance workshop
Equipment			
Assembly			
Order Type	PM02		Secondary Maintenance
Notification No.	**67648532**		R5 Tread-Recall RD88
Sort Field			
Planner Group	RW1		Wagons

Order Start Date	20.04.2026	**Order End Date**	20.04.2026
Priority	B	Medium Priority	
ABC Indicator (Criticality)	3	Class III - High	
Revisión	W7WG2617	RAIL 7 Mile Wagons 2026 Week 17	
Task List	A 127480 B6	**Plan**	
TL Critical Risk Assessed			

Order Long Text

L5 Tread-Must cut RD88
26.08.2025 04:00:19 UTC (ECP500PI-MPA) L5 Tread-Must cut RD88 B20020 5

Object List

FunctLocation	*3076WAG20020*	*.WSET*	Wheelset Assembly
Equipment			
Notification	67648532		L5 Tread-Must cut RD88
FunctLocation	*3076WAG20020*	*.WSET*	Wheelset Assembly
Equipment			
Notification	67648404		R3 Tread-Recall RD88
FunctLocation	*3076WAG20020*	*.WSET*	Wheelset Assembly
Equipment			
Notification	67648315		R5 Tread-Recall RD88

OpNo	**WorkCtr**	**Task Description**	**Duration**	**Actual**	
Functional Location			**No. People**		
Notification	**System Cond.**	**Operation Scheduled Start**	**Statutory**	**PSR**	**FSR**
Team Name		**Generic Field**			
0020	RMW11AE	88RD, Change out listed Wheelset	1,5 H	_____	
			2	_____	
	Off line	20.04.2026 00:00:00			

88RD, Change out listed Wheelset

Notes for the work being carried out on 88rd

",,These cars are not identified as overdue service cars
",,Remove and replace all wheel sets with a railbam score of 120 and above only
",,Visually inspect each couple knuckle or NDT if time Tac time allows, replace if cracked
" visual inspection of all components. If component is failed or past condemning unit wagon is to be carded for workshop. If compo
",,Work order to be completed by maintainer with all relevant tasks completed and singed off
",,Subsequent notifications are to be raised for knuckles if replaced. " Parts are to be booked to the work order, hrs updated, th

Document	RSM Orecar Work Order Time 10000393964 ZMP 000 00
Document	PM02 Wagon Inspection Without NDT 10000192317 ZMP 000 02
Document	PM02 extra additional worksheet - B 10000114057 ZMP 000 05
Document	RTIO-AM-0101618 Serv Info& Signoff Sheet 10000114021 ZMP 000 01

0030	RMW11BE	Transport, Wheelsets	0,5	H	_____
			1		_____
	Off line	20.04.2026 01:30:00			

0040	RMW	R3 Tread-Recall RD88	0,0	H	_____
3076WAG20020	.WSET	Wheelset Assembly	0		_____
67648404		20.04.2026 02:00:00			

0050	RMW	R5 Tread-Recall RD88	0,0	H	_____
3076WAG20020	.WSET	Wheelset Assembly	0		_____
67648315		20.04.2026 02:00:00			

Rio Tinto

MAINTENANCE ORDER

73188436

PICS Railway Seven Mile

Order Description	L5 Tread-Must cut RD88	
FunctLocation	3076WAG20020 .WSET	Wheelset Assembly
Productive Unit	3076WAG20020	Ore Wagon Pair B Type
Location	2RI0212030	Rolling stock maintenance workshop
Equipment		
Assembly		
Order Type	PM02	Secondary Maintenance
Notification No.	67648532	R5 Tread-Recall RD88
Sort Field		
Planner Group	RW1	Wagons

Information below is to be SAP Text

Comments

Name _____ Signature _____

Supervisor's Name _____ Supervisor's Signature _____

Downtime details:

Date Started:__/__/__ Time Started:__:__ Date Finished:__/__/__ Time Finished:__:__

RSM Orecar Work Order Time

Chronicle ID: 10000393964

Maintainer	Personnel Number	Work Date	Work Order Operation Number	Hours

Internal Use Only © 2024 Rio Tinto

PM02 Wagon Inspection Without NDT

Amendment record

This table identifies the changes to the document made for each new version.

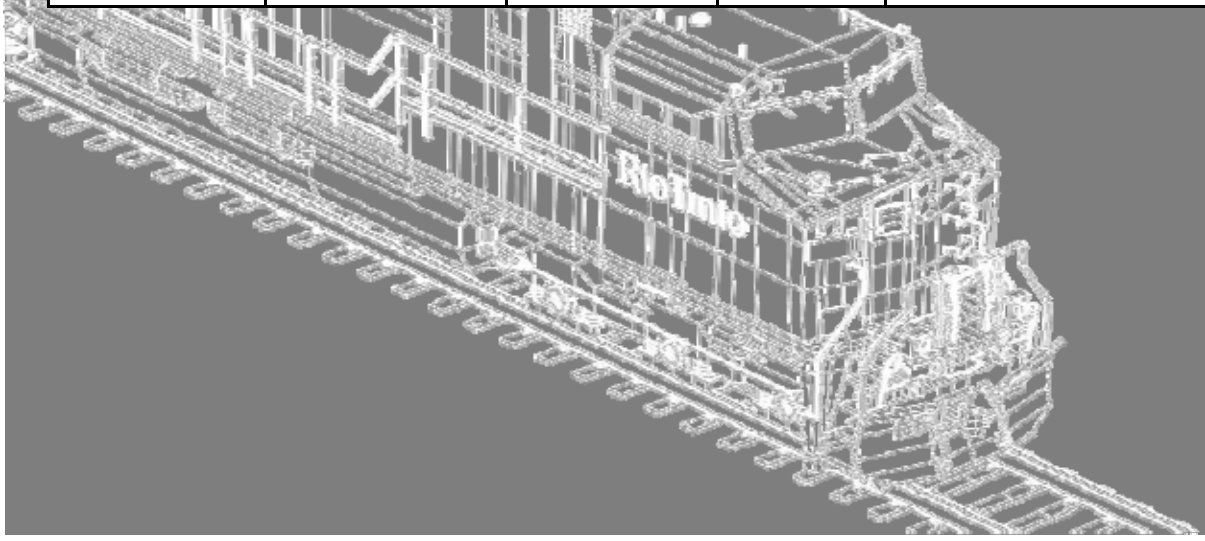
Docum ent Control and revision history

Docum e nt type: Managem ent Procedure

Version	Date	Author	Rev iewer	Approv ed	Approv ed for use from
1.0	04/02/2020	Mat Watkins	Declan Boy ham	May ur Kacha	04/02/2020
2.0	31/05/2023	Harsh Patel	Shoeb Ahmed	Matthew Wojas	31/05/2023

Am endm ent Record **Docum ent Area of Accountability: GM Railw ays Division**

Version & Date	Author	Change Management Number	Section Number	Modification Description
1.0 31-May-2023	H Patel	N/A	N/A	• Removed “Measure Rotary Collar” and included “Inspect Rotary Collar”



1. Purpose and Scope

This manual provides practical guidance and information about PM02 for Wagon Inspection.

- **Aim** — The aim of this task is to check and inspect equipment for defects. These checks are aimed at keeping the wagons in operable condition and catching potential breakdowns before they become actual breakdowns.
- **Instructions** — Perform all inspections with reference to the SWP & Rolling Stock Shop Standard
- **Isolations** — Isolations as per Ironsafe standards.
- **PPE Requirements** — PPE as per Ironsafe standards.
- **Tools& Equipment** — Tools and equipment to comply with Ironsafe standards.
- **Special Notes** — All maintainers shall tick all operations that they complete on the service sheet and sign off on the last page. Completed sheets are to be handed to the relevant Supervisor. If any General Inspection is found to be in need of repair the 'Repaired' box is to beticked.

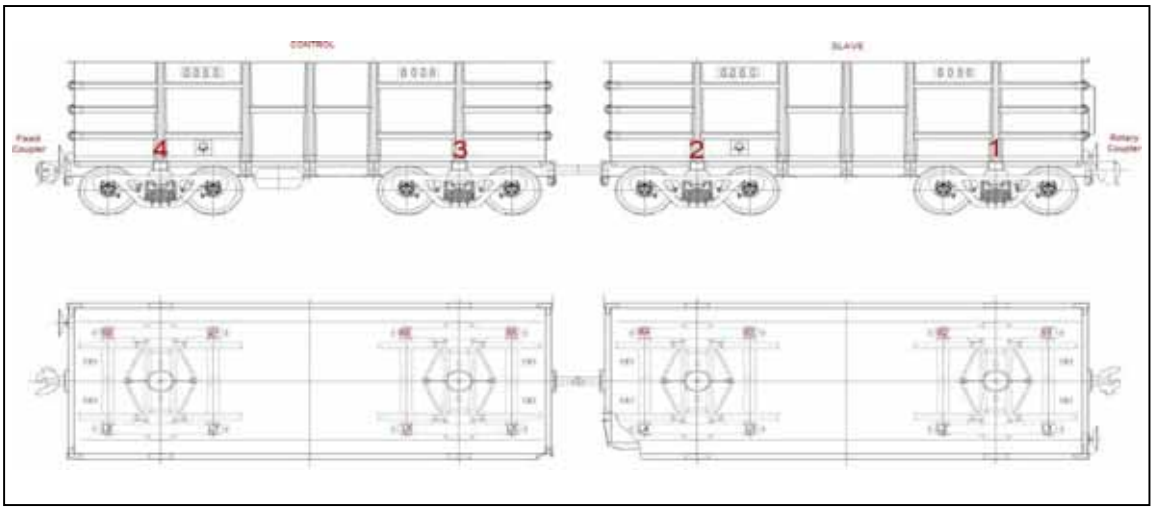
Brakes	Drawgear	Wheels	Bearings	ECP
Next Service Due				

Wagon Number	

TAKE FIVE TO BE COMPLETED PRIOR TO COMMENCING ANY TASK

Documentation Sign Off	Notification Follow Up Required	YES/NO
	Document Update Required	YES/NO
Name:	**Sign:**	**Date:**
Supervisor:	**Signature:**	**Date:**
Comments:		

Copyright Statement
© 2023 Rio Tinto Internal Use Only



8 Mile Inspection	Position				
Inspections	**1**	**2**	**3**	**4**	**SWP Ref:**
Visually inspect ACI tag.					
Visually inspect brake shoes and brake shoe keys.					
Visually inspect bearing keepers.					
Visually inspect inner & outer coil springs.					
Visually inspect stucki pads and cages.					
Measure bearing adapters on all replaced axles.					
Visually inspect brake hoses.					
Visually inspect brake rigging (slack adjusters, pins, brakes beams).					
Visually inspect bag hoses and ITC cables.					
Visually inspect hand brake wheels.					
Check operation of hand brakes.					
Visually inspect reservoir tank tap.					
Visually inspect wheel condition.					
Rotary and Fixed Coupler	**1**	**2**	**3**	**4**	**SWP Ref:**
Inspect knuckle, coupler and release rod operation.					
Inspect knuckle pins and spilt pin.					
Inspect fixed coupler ‘R’ clip.					
Rotary and Fixed Draft Gear	**1**	**2**	**3**	**4**	**SWP Ref:**
Inspect rotary collar.					
Support and Wear Plates	**1**	**2**	**3**	**4**	**SWP Ref:**
Inspect retainer plates nuts & bolts.					
Jacks – Wheel Replacement	**1**	**2**	**3**	**4**	**SWP Ref:**
Replace PM02 wheel sets.					
Record new wheel set numbers on SAP recording sheet.					
Replace bearing keeper nyloc nuts PM02 wheel sets.					
Check bearing adapter plates are sitting correctly with transdyne liners fitted.					

Copyright Statement
© 2023 Rio Tinto Internal Use Only

Service Information & Sign-Off

Control Car #		**Completion Date:**	

North		**South**	
Confirm		**TECO**	
Supervisor Name:		**Signature:**	

Release to Traffic Checklist	**Responsible**	**Name**
Ensure Bearing Keepers & locking tabs are in Place and secure & tabs bent over	Last maintainer to Work on wagon	
Ensure retainer plates are in place & secure	Last maintainer to Work on wagon	
Ensure that hand brake operates freely & is oiled	Last maintainer to Work on wagon	
Check that yellow cable tie is fitted	Last maintainer to Work on wagon	
Ensure the removal of the RED TAG in the Rat trap if good for shunting	Last maintainer to Work on wagon	
Ensure TSPB valve is in the open position	Last maintainer to Work on wagon	
Walk round wagon for general once over inspection	Last maintainer to Work on wagon	
Ensure correct Wheel information is captured on the Data sheet	Supervisors	

RioTinto

PM02/03 extra additional worksheet - B Series

Task 1	QTY	'B' Series Rotary Draft Gear TL 222842 – C3	Task 8	QTY	'B' Series Fixed Knuckle - TL 222842 - B5	Task 27	QTY	'B' Series Bag hose - TL 222842 - N9
40577311	1	YOKE;ROTARY;BRADKEN;AE270-2C	40577252	1	Knuckle (Bradken)	40577244	1	760mm Bag hose
40577314	1	COLLAR,BRADKEN 11318A4DWG	40577266	1	Knuckle pin (if required)	Task 28	QTY	'B' Series Branch Pipe Hose - TL 222842 - P2
40577241	1	PLATE;FRONT FOLLOWER;BRADKEN AE664	40577253	1	Locking Block (Bradken)	20536481	1	25NB Hose (1600mm)
40577239	1	Draft package miner SL-76 (new if required)	Task 9	QTY	B' Series Drawbar - TL 222842 - E3	Task 29	QTY	'B' Series RH end cock valve - TL 222842 - P0
40577276	2	RETAINING PLATE;COUPLER;HI T-002-215	40577238	1	Drawbar (Bradken)	20536500	1	COCK;END;B;RH;BRADKEN;KRV-125F
Task 2	QTY	'B' Series Rotary coupler - TL 222842 - E7	40577265	1	Fixed coupler pin	Task 30	QTY	'B' Series LH end cock valve - TL 222842 - P1
40577233	1	Rotary coupler (Bradken)	Task 10	QTY	'B' Series slack adjuster Pos 1 - TL 222842 - N0	20536499	1	COCK;END;B;LH;BRADKEN;KLV-125F
40577263	1	Rotary coupler pin (Bradken if required)	40577152	1	SLACK ADJUSTER,C1000 BOGIE BRAKE	Task 31	QTY	' B ' Series J1 Relay valve - TL 222842 - F8
Task 3	QTY	'B' Series Rotary Knuckle - TL 222842 - B6	Task 11	QTY	'B' Series slack adjuster Pos 2 - TL 222842 - N1	40577129	1	J1 Relay valve
40577252	1	Knuckle (Bradken)	Task 12	QTY	'B' Series slack adjuster Pos 3 - TL 222842 - N2	Task 32	QTY	'B' Series ELX-B Proportional valve - TL 222842 - D5
40577266	1	Knuckle pin (if required)	Task 13	QTY	'B' Series slack adjuster Pos 4 - TL 222842 - N3	20511366	1	ELX-B Proportional valve,40%
40577253	1	Locking Block (Bradken)	Task 14	QTY	'B' Series Brake beam Pos 1 - TL 222842 - H1	Task 33	QTY	B' Series AB-1 Check Valve - TL 222842 - F9
Task 4	QTY	'B' Series Fixed Draft Gear Pos 2 - TL 222842 - B9	40577148	1	Brake beam (TMX system, note position)	40577128	1	VALVE,CHECK;WABCO;563460
40577265	1	Fixed coupler pin	Task 15	QTY	'B' Series Brake beam Pos 2 - TL 222842 - H2	Task 34	QTY	'B' Series ABDX Service Portion - TL 222842 - A4
40577313	1	YOKE;FIXED	Task 16	QTY	'B' Series Brake beam Pos 3 - TL 222842 - H3	40577125	1	ABDX A,B & C Service Portion
40577240	1	FOLLOWER;FIXED COUPLER	Task 17	QTY	'B' Series Brake beam Pos 4 - TL 222842 - H4	Task 35	QTY	'B' Series ABDX Emergency Portion - TL 222842 - A0
20552473	1	PLATE,RETAINER;GUIDINGPIN (new if required)	Task 18	QTY	'B' Series Brake bucket Pos 1 - TL 222842 - H5	40748234	1	ABDX types A,B & C Emergency Portion
40577239	1	Draft package miner SL-76 (new if required)	20291408	1	Brake bucket (TMX system, note position)	Task 36	QTY	ECP
Task 5	QTY	'B' Series Fixed Draft Gear Pos 3 - TL 222842 - C0	Task 19	QTY	'B' Series Brake bucket Pos 2 - TL 222842 - H6	20288197	1	CCD, Door only (28434P) **TL 222842 - E1**
40577265	1	Fixed coupler pin	Task 20	QTY	'B' Series Brake bucket Pos 3 - TL 222842 - H7	20494938	1	CID, Board only (27718P) **TL 222842 - E2**
40577313	1	YOKE;FIXED	Task 21	QTY	'B' Series Brake bucket Pos 4 - TL 222842 - H8	20494941	1	CID box (Bottom section-33221P) **TL 222842 - E4**
40577240	1	FOLLOWER;FIXED COUPLER	Task 22	QTY	'B' Series Brake bucket hose TL 222842 - N6	20776034	1	BATTERY;POWENESO;LFP12.8-4 LIFEPO4 **TL 222842-G3**
Task 6	QTY	'B' Series Fixed Draft Gear Pos 4 - TL 222842 - C1	40577138	1	Brake bucket hose 36" long (note position)	20225537	1	SCD, Intercar cable (24315P) **TL 222842 - G4**
40577265	1	Fixed coupler pin	Task 23	QTY	'B' Series Control line air hose - TL 222842 - N7	20697411	1	Intercar cable Twist lock 36 inch(35800P) **TL 222842 - G5**
40577313	1	YOKE;FIXED	40577132	1	3/4' Brake hose (in-between ore cars)	20373527	1	CCID to manifold cable (28026-2) **TL 222842 - G6**
40577240	1	FOLLOWER;FIXED COUPLER	Task 24	QTY	'B' Series Train line air hose - TL 222842 - N8	20536384	1	Train line Armoured Cable (5100mm) **TL 222842 - P9**
20552473	1	PLATE,RETAINER;GUIDINGPIN (new if required)	40577133	1	1 3/8' Train line air hose (in-between ore cars)	20536385	1	Train line Armoured Cable (5080mm) **TL 222842 – Q0**
40577239	1	Draft package miner SL-76 (new if required)	Task 25	QTY	'B' Series Supply Reservoir Hose TL 222842 - P7	20225552	1	Junction box (slave car- 28792P)
40577219	2	WEAR PLATE;STRIKER ASSY;FRONT STOPS	20536493	1	20NB Hose (520mm)	20283942	1	ECP Overlay manifold (666435) **TL 222842 - D1**
Task 7	QTY	'B' Series Fixed coupler - TL 222842 - F1	Task 26	QTY	'B' Series Combined Reservoir Hose - TL 222842 - P8	PM02 WAGON INSPECTION TL - 222842 - B1		
40577232	1	Fixed coupler (Bradken)	20536496	2	20NB Hose (1330mm)			
40577265	1	Fixed coupler pin						

Task	Component	Name	Date	SAP	Hrs	Maintenance Order	Opp	Reset	Comments

FDMS File RTIO-AM-0179818 **Changed by: Jason Swalling** **Reviewed by: Billie-Jo O'Brien** **Approved by: Joshua Norman** **Date Implemented: 04/11/2025**

Rio Tinto

MAINTENANCE ORDER

74377116

PICS Railway Seven Mile

Order Description	Install label on the 98L Combined Tanks	
FunctLocation	3076WAG20020 .ABRK.ARCR	Air Receiver: Combined Reservoir
Productive Unit	3076WAG20020	Ore Wagon Pair B Type
Location	2RI0212030	Rolling stock maintenance workshop
Equipment		
Assembly		
Order Type	PM02	Secondary Maintenance
Notification No.	**72358547**	Install label on the 98L Combined Tanks
Sort Field		
Planner Group	RW1	Wagons

Order Start Date	20.04.2026	**Order End Date**	20.04.2026
Priority	C	Low Priority	
ABC Indicator (Criticality)	3	Class III - High	
Revisión	W7WG2617	RAIL 7 Mile Wagons 2026 Week 17	
Task List	A 371663 1	**Plan**	2225427
TL Critical Risk Assessed			

Order Long Text

Install label on the 98L Combined Tanks

Install label on the 98L combined Tanks

Install the related label on the 98L tank on all wagons from Tag 1 to 51 as shown here. Use the Caulking gun with Sikaflex to apply label.

OpNo	**WorkCtr**	**Task Description**	**Duration**	**Actual**	
Functional Location			**No. People**		
Notification	**System Cond.**	**Operation Scheduled Start**	**Statutory**	**PSR**	**FSR**
Team Name		**Generic Field**			
0010	RMW11AC	Install label on the 98L combined Tanks	0,0 H	_____	
3076WAG20020	.ABRK.ARCR	Air Receiver: Combined Reservoir	0	____	
72358547	Off line	20.04.2026 00:00:00			

Install label on the 98L combined Tanks

Install the related label on the 98L tank on all wagons from Tag 1 to 51 as shown here. Use the Caulking gun with Sikaflex to apply

Document Auxilliary Tank PV Labels Tag 1 to 51 10000533229 ZMP 000 00

Rio Tinto

MAINTENANCE ORDER

74377116

PICS Railway Seven Mile

Order Description	Install label on the 98L Combined Tanks	
FunctLocation	3076WAG20020 .ABRK.ARCR	Air Receiver: Combined Reservoir
Productive Unit	3076WAG20020	Ore Wagon Pair B Type
Location	2RI0212030	Rolling stock maintenance workshop
Equipment		
Assembly		
Order Type	PM02	Secondary Maintenance
Notification No.	72358547	Install label on the 98L Combined Tanks
Sort Field		
Planner Group	RW1	Wagons

Information below is to be SAP Text

Comments

Name _____ Signature _____

Supervisor's Name _____ Supervisor's Signature _____

Downtime details:

Date Started:__/__/__ Time Started:__:__ Date Finished:__/__/__ Time Finished:__:__

PV Labels Tag 1 to 51 - Auxiliary Reservoir Tank.

Item 1: Tag 1 to 51 wagons only (non-blue wagons)

Install label as shown on the auxiliary tank.

RE-RATED BY: RIO TINTO IRON ORE
VERIFIED BY: PACIFIC ENGINEERING
DESIGN PRESSURE: 620.5kPa AS1210-3
DESIGN TEMPERATURE: 120°C
TEST PRESSURE: 1310kPa
SERVICE FACTOR: 0.33 - CLAUSE 2.2.5c(1)
REGISTRATION NO.: N/A
HAZARD LEVEL: E(EMER) E(AUX)





Rio Tinto

MAINTENANCE ORDER

74384529

PICS Railway Seven Mile

Order Description	Install Label on the 165L Supply Tank	
FunctLocation	3076WAG20020 .ABRK.ARSR	Air Receiver: Supply Reservoir
Productive Unit	3076WAG20020	Ore Wagon Pair B Type
Location	2RI0212030	Rolling stock maintenance workshop
Equipment		
Assembly		
Order Type	PM02	Secondary Maintenance
Notification No.	**72365912**	Install Label on the 165L Supply Tank
Sort Field		
Planner Group	RW1	Wagons

Order Start Date	20.04.2026	**Order End Date**	20.04.2026
Priority	C	Low Priority	
ABC Indicator (Criticality)	3	Class III - High	
Revisión	W7WG2617	RAIL 7 Mile Wagons 2026 Week 17	
Task List	A 371663 2	**Plan**	2225438
TL Critical Risk Assessed			

Order Long Text

Install Label on the 165L Supply Tank

Install Label on the 165L supply tank

Install the related label on the 165L tank on all wagons from Tag 1 to 51 as shown here. Use the Caulking gun with Sikaflex to apply label.

OpNo	WorkCtr	Task Description	Duration	Actual	
Functional Location			**No. People**		
Notification	**System Cond.**	**Operation Scheduled Start**	**Statutory**	**PSR**	**FSR**
Team Name		**Generic Field**			
0010	RMW11AC	Install Label on the 165L supply tank	0,0 H	_____	
3076WAG20020	.ABRK.ARSR	Air Receiver: Supply Reservoir	0	_____	
72365912	Off line	20.04.2026 00:00:00			

Install Label on the 165L supply tank

Install the related label on the 165L tank on all wagons from Tag 1 to 51 as shown here. Use the Caulking gun with Sikaflex to apply

Document Main Supply Resevoir PV Labels Tag 1 to 10000533260 ZMP 000 00

Rio Tinto

MAINTENANCE ORDER

74384529

PICS Railway Seven Mile

Order Description	Install Label on the 165L Supply Tank		
FunctLocation	3076WAG20020	.ABRK.ARSR	Air Receiver: Supply Reservoir
Productive Unit	3076WAG20020		Ore Wagon Pair B Type
Location	2RI0212030		Rolling stock maintenance workshop
Equipment			
Assembly			
Order Type	PM02		Secondary Maintenance
Notification No.	72365912		Install Label on the 165L Supply Tank
Sort Field			
Planner Group	RW1		Wagons

Information below is to be SAP Text

Comments

Name _____ Signature _____

Supervisor's Name _____ Supervisor's Signature _____

Downtime details:

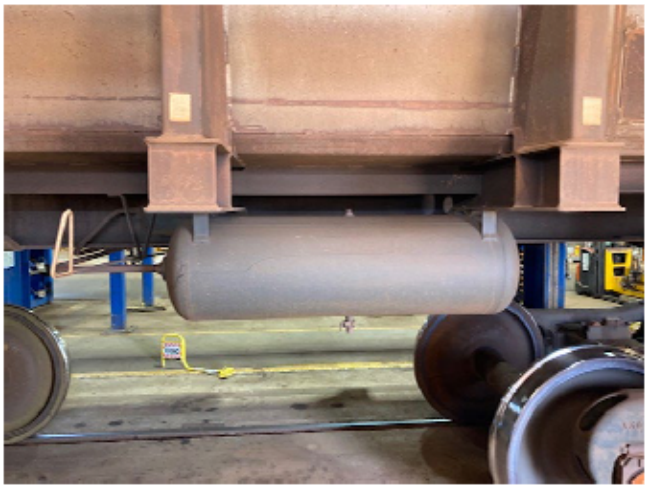
Date Started:__/__/__ Time Started:__:__ Date Finished:__/__/__ Time Finished:__:__

PV Labels Tag 1 to 51 – Main Supply Reservoir Tank.

Item 1: Tag 1 to 51 wagons only (non-blue wagons)

Install label as shown on the main supply reservoir tank.

RE-RATED BY: RIO TINTO IRON ORE
VERIFIED BY: PACIFIC ENGINEERING
DESIGN PRESSURE: 620.5kPa AS1210-3
DESIGN TEMPERATURE: 120°C
TEST PRESSURE: 1586kPa
SERVICE FACTOR: 0.33 - CLAUSE 2.2.5c(1)
REGISTRATION No.: WAP27675
HAZARD LEVEL: D





Rio Tinto

MAINTENANCE ORDER

74396212

PICS Railway Seven Mile

Order Description	Brake System Failed		
FunctLocation	3076WAG20020	.ABRK	Brake System
Productive Unit	3076WAG20020		Ore Wagon Pair B Type
Location	2RI0212030		Rolling stock maintenance workshop
Equipment			
Assembly			
Order Type	PM02		Secondary Maintenance
Notification No.	**72412119**		very minor air leak from ECP manifold
Sort Field			
Planner Group	RW1		Wagons

Order Start Date	20.04.2026	**Order End Date**	20.04.2026
Priority	B	Medium Priority	
ABC Indicator (Criticality)	3	Class III - High	
Revisión	W7WG2617	RAIL 7 Mile Wagons 2026 Week 17	
Task List	A 257136 1	**Plan**	
TL Critical Risk Assessed			

Order Long Text

Brake System Failed
13.04.2026 06:14:14 UTC (JBELL011) Phone + 61 (425705612)
This wagon has been flagged by the HBD as having bogie cold wheels.
Please inspect and circle the identified failure modes and the position:

POS1 POS2 POS3 POS4

1. slack adjuster over-travel item broken
2. slack adjuster travel adjustment required
3. Brake Cylinder leaking
4. Brake Cylinder Piston Travel Adjustment Required
5. Leaking or cracked hoses or pipes
6. End cocks leaking
7. Brake Shoes

other failure modes:

_____
_____
_____

PLEASE RECORD ALL IDENTIFIED FAILURE MODES ON THIS NOTIFICATION

Rio Tinto

MAINTENANCE ORDER

74396212

PICS Railway Seven Mile

and alert your supervisor so that failure modes are communicated to the Reliability Team.

Object List

FunctLocation	*3076WAG20020*	*.ABRK*	Brake System
Equipment			
Notification	72412119		Brake System Failed
FunctLocation	*3076WAG20020*		Ore Wagon Pair B Type
Equipment			
Notification	72187381		very minor air leak from ECP manifold

OpNo	WorkCtr	Task Description	Duration	Actual
Functional Location			**No. People**	
Notification	**System Cond.**	**Operation Scheduled Start**	**Statutory**	**PSR FSR**
Team Name		**Generic Field**		
0010	RMW11AE	86/87RD PM02 Wagon Inspection	3,0 H	_____
			1	_____
	Off line	20.04.2026 00:00:00		
	Document	RSM Orecar Work Order Time 10000393964 ZMP 000 00		
	Document	PM02 Wagon Inspection Without NDT 10000192317 ZMP 000 02		
	Document	PM02 extra additional worksheet - B 10000114057 ZMP 000 05		
	Document	RTIO-AM-0180519 - Pneumatic Single Car B 10000114019 ZMP 000 09		
	Document	RTIO-AM-0101618 Serv Info& Signoff Sheet 10000114021 ZMP 000 01		
0020	RMW31AE	ECP Brake Test & Service	1,0 H	_____
			1	_____
	Off line	20.04.2026 03:00:00		
	Document	RTIO-AM-0179836 ECP Brake Test & Service 10000114018 ZMP 000 07		

Rio Tinto

MAINTENANCE ORDER

74396212

PICS Railway Seven Mile

Order Description	Brake System Failed		
FunctLocation	3076WAG20020	.ABRK	Brake System
Productive Unit	3076WAG20020		Ore Wagon Pair B Type
Location	2RI0212030		Rolling stock maintenance workshop
Equipment			
Assembly			
Order Type	PM02		Secondary Maintenance
Notification No.	72412119		very minor air leak from ECP manifold
Sort Field			
Planner Group	RW1		Wagons

Information below is to be SAP Text

Comments

Name _____ Signature _____

Supervisor's Name _____ Supervisor's Signature _____

Downtime details:

Date Started:__/__/__ Time Started:__:__ Date Finished:__/__/__ Time Finished:__:__

RSM Orecar Work Order Time

Chronicle ID: 10000393964

Maintainer	Personnel Number	Work Date	Work Order Operation Number	Hours
Pragash Murugesan	9444361	23/04/2026	213456789	0.5

Internal Use Only © 2024 Rio Tinto

PM02 Wagon Inspection Without NDT

Amendment record

This table identifies the changes to the document made for each new version.

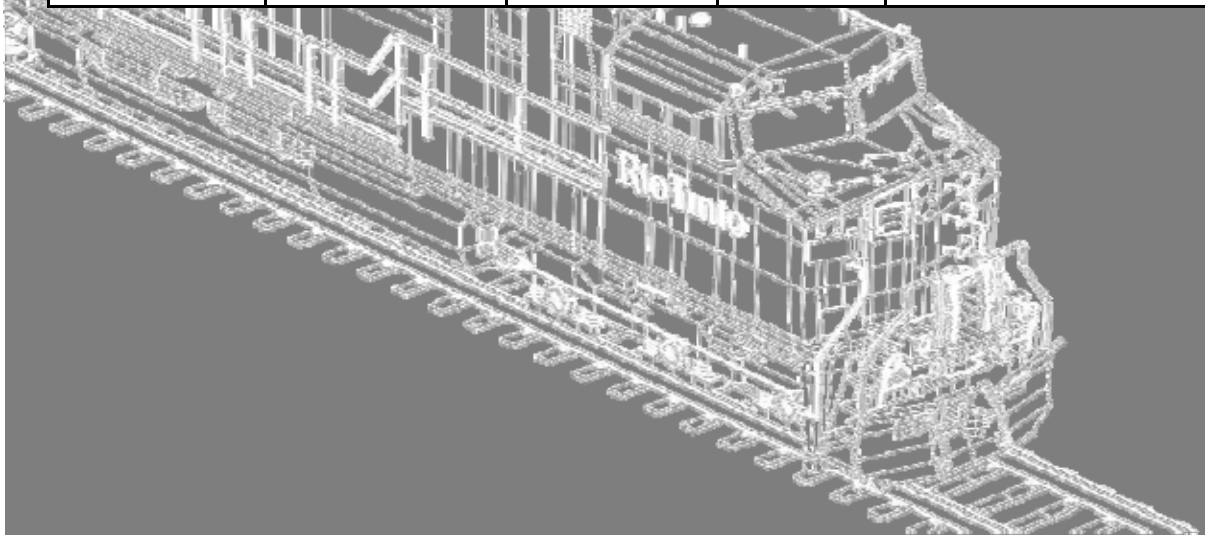
Docum ent Control and revision history

Docum e nt type: Managem ent Procedure

Version	Date	Author	Rev iewer	Approv ed	Approv ed for use from
1.0	04/02/2020	Mat Watkins	Declan Boy ham	May ur Kacha	04/02/2020
2.0	31/05/2023	Harsh Patel	Shoeb Ahmed	Matthew Wojas	31/05/2023

Am endm ent Record **Docum ent Area of Accountability: GM Railw ays Division**

Version & Date	Author	Change Management Number	Section Number	Modification Description
1.0 31-May-2023	H Patel	N/A	N/A	• Removed “Measure Rotary Collar” and included “Inspect Rotary Collar”



1. Purpose and Scope

This manual provides practical guidance and information about PM02 for Wagon Inspection.

- **Aim** — The aim of this task is to check and inspect equipment for defects. These checks are aimed at keeping the wagons in operable condition and catching potential breakdowns before they become actual breakdowns.
- **Instructions** — Perform all inspections with reference to the SWP & Rolling Stock Shop Standard
- **Isolations** — Isolations as per Ironsafe standards.
- **PPE Requirements** — PPE as per Ironsafe standards.
- **Tools& Equipment** — Tools and equipment to comply with Ironsafe standards.
- **Special Notes** — All maintainers shall tick all operations that they complete on the service sheet and sign off on the last page. Completed sheets are to be handed to the relevant Supervisor. If any General Inspection is found to be in need of repair the 'Repaired' box is to beticked.

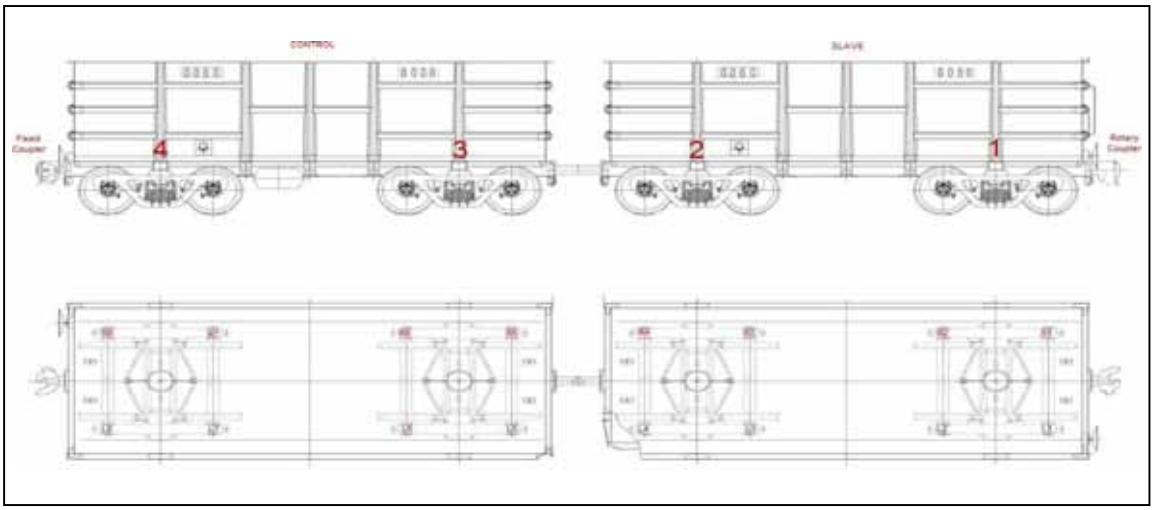
Brakes	Drawgear	Wheels	Bearings	ECP
Next Service Due				

Wagon Number	

TAKE FIVE TO BE COMPLETED PRIOR TO COMMENCING ANY TASK

Documentation Sign Off	Notification Follow Up Required	YES/NO
	Document Update Required	YES/NO
Name:	**Sign:**	**Date:**
Supervisor:	**Signature:**	**Date:**
Comments:		

Copyright Statement
© 2023 Rio Tinto Internal Use Only



8 Mile Inspection	Position				
Inspections	**1**	**2**	**3**	**4**	**SWP Ref:**
Visually inspect ACI tag.					
Visually inspect brake shoes and brake shoe keys.					
Visually inspect bearing keepers.					
Visually inspect inner & outer coil springs.					
Visually inspect stucki pads and cages.					
Measure bearing adapters on all replaced axles.					
Visually inspect brake hoses.					
Visually inspect brake rigging (slack adjusters, pins, brakes beams).					
Visually inspect bag hoses and ITC cables.					
Visually inspect hand brake wheels.					
Check operation of hand brakes.					
Visually inspect reservoir tank tap.					
Visually inspect wheel condition.					
Rotary and Fixed Coupler	**1**	**2**	**3**	**4**	**SWP Ref:**
Inspect knuckle, coupler and release rod operation.					
Inspect knuckle pins and spilt pin.					
Inspect fixed coupler ‘R’ clip.					
Rotary and Fixed Draft Gear	**1**	**2**	**3**	**4**	**SWP Ref:**
Inspect rotary collar.					
Support and Wear Plates	**1**	**2**	**3**	**4**	**SWP Ref:**
Inspect retainer plates nuts & bolts.					
Jacks – Wheel Replacement	**1**	**2**	**3**	**4**	**SWP Ref:**
Replace PM02 wheel sets.					
Record new wheel set numbers on SAP recording sheet.					
Replace bearing keeper nyloc nuts PM02 wheel sets.					
Check bearing adapter plates are sitting correctly with transdyne liners fitted.					

Copyright Statement
© 2023 Rio Tinto Internal Use Only

RioTinto

PM02/03 extra additional worksheet - B Series

Task 1	QTY	'B' Series Rotary Draft Gear TL 222842 – C3	Task 8	QTY	'B' Series Fixed Knuckle - TL 222842 - B5	Task 27	QTY	'B' Series Bag hose - TL 222842 - N9
40577311	1	YOKE;ROTARY;BRADKEN;AE270-2C	40577252	1	Knuckle (Bradken)	40577244	1	760mm Bag hose
40577314	1	COLLAR,BRADKEN 11318A4DWG	40577266	1	Knuckle pin (if required)	Task 28	QTY	'B' Series Branch Pipe Hose - TL 222842 - P2
40577241	1	PLATE;FRONT FOLLOWER;BRADKEN AE664	40577253	1	Locking Block (Bradken)	20536481	1	25NB Hose (1600mm)
40577239	1	Draft package miner SL-76 (new if required)	Task 9	QTY	B' Series Drawbar - TL 222842 - E3	Task 29	QTY	'B' Series RH end cock valve - TL 222842 - P0
40577276	2	RETAINING PLATE;COUPLER;HI T-002-215	40577238	1	Drawbar (Bradken)	20536500	1	COCK;END;B;RH;BRADKEN;KRV-125F
Task 2	QTY	'B' Series Rotary coupler - TL 222842 - E7	40577265	1	Fixed coupler pin	Task 30	QTY	'B' Series LH end cock valve - TL 222842 - P1
40577233	1	Rotary coupler (Bradken)	Task 10	QTY	'B' Series slack adjuster Pos 1 - TL 222842 - N0	20536499	1	COCK;END;B;LH;BRADKEN;KLV-125F
40577263	1	Rotary coupler pin (Bradken if required)	40577152	1	SLACK ADJUSTER,C1000 BOGIE BRAKE	Task 31	QTY	' B ' Series J1 Relay valve - TL 222842 - F8
Task 3	QTY	'B' Series Rotary Knuckle - TL 222842 - B6	Task 11	QTY	'B' Series slack adjuster Pos 2 - TL 222842 - N1	40577129	1	J1 Relay valve
40577252	1	Knuckle (Bradken)	Task 12	QTY	'B' Series slack adjuster Pos 3 - TL 222842 - N2	Task 32	QTY	'B' Series ELX-B Proportional valve - TL 222842 - D5
40577266	1	Knuckle pin (if required)	Task 13	QTY	'B' Series slack adjuster Pos 4 - TL 222842 - N3	20511366	1	ELX-B Proportional valve,40%
40577253	1	Locking Block (Bradken)	Task 14	QTY	'B' Series Brake beam Pos 1 - TL 222842 - H1	Task 33	QTY	B' Series AB-1 Check Valve - TL 222842 - F9
Task 4	QTY	'B' Series Fixed Draft Gear Pos 2 - TL 222842 - B9	40577148	1	Brake beam (TMX system, note position)	40577128	1	VALVE,CHECK;WABCO;563460
40577265	1	Fixed coupler pin	Task 15	QTY	'B' Series Brake beam Pos 2 - TL 222842 - H2	Task 34	QTY	'B' Series ABDX Service Portion - TL 222842 - A4
40577313	1	YOKE;FIXED	Task 16	QTY	'B' Series Brake beam Pos 3 - TL 222842 - H3	40577125	1	ABDX A,B & C Service Portion
40577240	1	FOLLOWER;FIXED COUPLER	Task 17	QTY	'B' Series Brake beam Pos 4 - TL 222842 - H4	Task 35	QTY	'B' Series ABDX Emergency Portion - TL 222842 - A0
20552473	1	PLATE,RETAINER;GUIDINGPIN (new if required)	Task 18	QTY	'B' Series Brake bucket Pos 1 - TL 222842 - H5	40748234	1	ABDX types A,B & C Emergency Portion
40577239	1	Draft package miner SL-76 (new if required)	20291408	1	Brake bucket (TMX system, note position)	Task 36	QTY	ECP
Task 5	QTY	'B' Series Fixed Draft Gear Pos 3 - TL 222842 - C0	Task 19	QTY	'B' Series Brake bucket Pos 2 - TL 222842 - H6	20288197	1	CCD, Door only (28434P) **TL 222842 - E1**
40577265	1	Fixed coupler pin	Task 20	QTY	'B' Series Brake bucket Pos 3 - TL 222842 - H7	20494938	1	CID, Board only (27718P) **TL 222842 - E2**
40577313	1	YOKE;FIXED	Task 21	QTY	'B' Series Brake bucket Pos 4 - TL 222842 - H8	20494941	1	CID box (Bottom section-33221P) **TL 222842 - E4**
40577240	1	FOLLOWER;FIXED COUPLER	Task 22	QTY	'B' Series Brake bucket hose TL 222842 - N6	20776034	1	BATTERY;POWENESO;LFP12.8-4 LIFEPO4 **TL 222842-G3**
Task 6	QTY	'B' Series Fixed Draft Gear Pos 4 - TL 222842 - C1	40577138	1	Brake bucket hose 36" long (note position)	20225537	1	SCD, Intercar cable (24315P) **TL 222842 - G4**
40577265	1	Fixed coupler pin	Task 23	QTY	'B' Series Control line air hose - TL 222842 - N7	20697411	1	Intercar cable Twist lock 36 inch(35800P) **TL 222842 - G5**
40577313	1	YOKE;FIXED	40577132	1	3/4' Brake hose (in-between ore cars)	20373527	1	CCID to manifold cable (28026-2) **TL 222842 - G6**
40577240	1	FOLLOWER;FIXED COUPLER	Task 24	QTY	'B' Series Train line air hose - TL 222842 - N8	20536384	1	Train line Armoured Cable (5100mm) **TL 222842 - P9**
20552473	1	PLATE,RETAINER;GUIDINGPIN (new if required)	40577133	1	1 3/8' Train line air hose (in-between ore cars)	20536385	1	Train line Armoured Cable (5080mm) **TL 222842 – Q0**
40577239	1	Draft package miner SL-76 (new if required)	Task 25	QTY	'B' Series Supply Reservoir Hose TL 222842 - P7	20225552	1	Junction box (slave car- 28792P)
40577219	2	WEAR PLATE;STRIKER ASSY;FRONT STOPS	20536493	1	20NB Hose (520mm)	20283942	1	ECP Overlay manifold (666435) **TL 222842 - D1**
Task 7	QTY	'B' Series Fixed coupler - TL 222842 - F1	Task 26	QTY	'B' Series Combined Reservoir Hose - TL 222842 - P8	PM02 WAGON INSPECTION TL - 222842 - B1		
40577232	1	Fixed coupler (Bradken)	20536496	2	20NB Hose (1330mm)			
40577265	1	Fixed coupler pin						

Task	Component	Name	Date	SAP	Hrs	Maintenance Order	Opp	Reset	Comments

FDMS File RTIO-AM-0179818 **Changed by: Jason Swalling** **Reviewed by: Billie-Jo O'Brien** **Approved by: Joshua Norman** **Date Implemented: 04/11/2025**

RTIO-AM-0180519

Pneumatic Single Car Brake Test Sign-off sheet for Pool fleet wagons

Amendment record

This table identifies the changes to the document made for each new version.

Document Control and revision history **Document type: Management Procedure**

RTIO-AM-0810519

Version	Date	Author	Reviewer	Approved by	Approved for use from
2.4.1	25 June 2015	Stephen Winter	Brooke Lane	Way ne Moss	03 June 2015
2.4.2	20 Dec 2016	Foha Rafiq	Ian Harding	John R. McArthur	06 September 2017
2.4.3	01 Mar 2018	Tatiana Joannides	John R. McArthur	Allan Liu	16 March 2018
10.0	15 Nov 2018	Wynand Jordaan	John R. McArthur	Allan Liu	31 Nov ember 2018
10.1	01 Oct 2019	Wynand Jordaan	Stephen Karay	John R. McArthur	01 October 2018
10.2	27 April 2020	Darren Brehmer	Stephen Karay	Declan Boy ham	27 April 2020
Issue 13	15 July 2020	Wynand Jordaan	Darren Brehmer	Declan Boy ham	1 August 2020
Issue 15	1 Aug 2022	Wynand Jordaan	Matthew Wojas	Declan Boyham	18 August 2020
Issue 16	22 Feb 2023	Richard Bosworth, Wynand Jordaan, Ricco Dharmawan	Matthew Wojas	Declan Boyham	1 March 2023
Issue 17	09/05/2024	Wynand Jordaan	Andrew Stevens	Matthew Wojas	09/05/2024
Issue 18	28/11/2025	Wynand Jordaan	Sheperd Murape	Billie-Jo O'Brien	01/01/2026

Amendment Record **Document Area of Accountability: GM Railw ays Division**

Version & Date		Author	Change Management Number	Section Number	Modification Description
2.4.2	20 Dec 2016	Foha Rafiq	43621		Updated Empty/Load Value Testing to include specifics for wagon series. Reviewed Standard RTIO-LO-0001846 and included in Reference.
2.4.3	01 Mar 2018	Tatiana Joannides		7.6	Slack adjuster travel specifications.
10.0	31 Nov 2018	Wynand Jordaan		All	Reference and version lock this document to RTIO-LO-0001846 the Single Car Brake test standard
10.1	01 Oct 2019	Wynand Jordaan		7.3	Increased downstream brake cylinder leakage to 3 psi
10.2	27 April 2020	Darren Brehmer	62484	7.6	Push rod length v s AAR wheel size
Issue 13	15 July 2020	Wynand Jordaan		7.3	Adjust BC Leak tolerances
Issue 15	1 Aug 2022	Wynand Jordaan	82060		New BC Leak tolerances
Issue 16	22 Feb 2023	Richard Bosworth, Wynand Jordaan		Various	Pressure conversions updates
Issue 17	09/05/2024	Wynand Jordaan	103365	Various	Revision updated to align with Standard RTIO-LO-0001846
Issue 18	28/11/2025	Wynand Jordaan	131239	Various	Additional info on TSPB pre checks Brake Cylinder Leak test – additional requirement for RTCs Removal of using the ECP box for Hipot testing New TSPB reporting marks for ECP SA wagons – excluding EOTC's

1. Purpose and Scope

This document provides Sign-off on the Pneumatic Brake Test & Service for Ore Cars.

This document must be used in conjunction with the Single Car Brake test standard RTIO-LO-0001846.

This document will serve as a written record for the Brake test performed using the Rio Tinto Single Car Brake test standard – RTIO-LO-0001846

1. NOTE:

The revision of this Document must at all times match the revision of RTIO-LO-0001846, and can never be amended in isolation. RTIO-LO-0001846 controls the revision of this document.

- **Aim** — The aim of this task is to check, inspect, repair and maintain equipment for safety and reliability. These inspections are aimed at keeping the ore cars in operable condition and catching potential breakdowns before they become actual breakdowns.
- **Instructions** — Defects are to be recorded and relevant comments listed.
- **Isolations** — Isolations as per Ironsafe standards.
- **PPE Requirements** — PPE as per Ironsafe standards.
- **Tools & Equipment** — Tools and equipment to comply with Ironsafe standards.
- **Special Notes** — All testers shall tick all operations that they complete on the service sheet and sign off on the last page. Completed sheets are to be handed to the relevant site supervisor

Reference Documents	**FDMS No:**
1. Single Car Brake Test Standard	RTIO-LO-0001846

Internal Use Only
© 2023 Rio Tinto

Ref	Pneumatic Testing		
Note: The complete brake test must be re-done if any of the components are replaced during the inspection / testing process			
7.1 & 7.2	**Preliminary Procedure and System Leakage test**	**Pass/Fail**	**Value**
	Replace both glad hand seals, Inspect the hose condition and orientation, Inspect the split pins are not worn and clevis pins are in the correct orientation, Test the hand brake operation. Application and release. Note: Wagons 31721 to 31780 have no handbrake installed.		
	System charged to 620kpa (90psi), Flowrator seize to rise. Movehandle to lap (3) Maximum allowed leakage in 2 minutes 3kPa (0.5psi)		
7.3	**Min Application test**		
	Did an audible exhaust come from the service portion?		
	Reduce brake pipe pressure to 552-586kPa (80-85psi). Allow it to stabilize and make sure it holds for 3 minutes.		
	Confirm brakes apply and both brake cylinder gauges indicate above 150 to 170kpa (22 to 25psi).		
7.4	**Slow Release test**		
	Move handle to slow release. Confirm BC pressure reach 0 kPa within 45 seconds		
7.5	**Service Stability**		
	Service stability check has not caused an emergency application		
7.6	**Piston Travel and Rigging – No adjustment required**		
	Ore cars fitted with **TMX-** Indicator within the notch. Ore cars with **TMB60** (KB wagons)- Indicator between 45-57mm		
	Confirm the correct Pushrod length - Wheel size 16-26 AAR: Push Rod Length 855 – 860 mm Wheel size 27-34 AAR: Push Rod Length 825 – 830 mm Wheel size 35-44 AAR: Push Rod Length 785 – 790 mm		
	Check that brake beams and push rods do not bind or foul on the bogie frame.		
7.7	**Piston Travel and Rigging – Adjustment required**		
	Piston travel have been readjusted to 2" ± ¼" (51 ± 6mm)	Bogie 1	
		Bogie 2	
		Bogie 3	
		Bogie 4	
7.8	**Slack adjuster test**		
	Confirm all 4 Slack adjusters are working		
7.11	**Emergency Application test**		
	Reduce BP to 275kPa (40psi) using position 5, then move handle to position 3. This must not cause an emergency application.		
	Open dump test cock quickly. Brake pipe reduces to zero and brake cylinder should read above 503kPa (73psi). Check brakes are applied on all wheels.		BP: BC:
	Close dump test cock and wait 3 minutes. Re-open dump test cock. No air should be vented.		
7.12	**Brake Cylinder leak test**		
	Upstream measurement taken at ELXB		
	Maximum permissible Leak rate is 7kPa/min with BC greater than 500kpa		
For **EOTC**	Maximum permissible Leak rate is 1.1kPa/min with BC greater than 500kpa		
For **Rail Tank Cars**	Maximum permissible Leak rate is 1 kPa/min with BC greater than 500kpa		
	Down-stream measurement taken at BC point over drawbar		
	Close J1 Isolation cock		
	Pull release rod to release BC – Note ELXB arm retracting		

Internal Use Only
© 2023 Rio Tinto

Printed Copies are Uncontrolled

General Business / Affaires générales

	Maximum permissible Leak rate is 7kPa/min with BC greater than 500kpa		
For **EOTC**	Maximum permissible Leak rate is 1.1kPa/min with BC greater than 500kpa		
7.13	**Release After Emergency**		
	Move handle to position 1 (release) and charge brake pipe to 193kPa (28psi), immediately move handle to position 1 and wait 1 minute. BP should rise to 220-350kPa (32-51psi).		
	Recharge the BP to 620kPa (90psi). Check brakes released.		
7.14	**Manual Release After Emergency**		
	Open dump test cock to create an emergency application. After 90 seconds, pull brake cylinder release rod and check brakes go back to release position and stay there.		
	Close dump test cock and re-charge to 620kPa (90psi). Brakes should remain released.		
7.15	**Empty/Load Valve Test**		
7.15.1	**S, C, J series with ELX-B (50%)**		
	Recharge BP and allow to stabilize. Reduce brake pipe to 448kPa (65psi). Confirm 5-6mm gap to the top of the bogie frame and valvearm and Empty indicator is protruding and brake cylinder reading 193-214kPa (28-31psi) on the gauge		BP: Gap: BC:
	Recharge BP, install 20mm block under valve arm. Reduce BP pressure to 448kPa (65psi). Arm contacting block, valve empty indicator not protruding and brake cylinder reading 400-427kPa (57-62psi) on the gauge		BP: Gap: BC:
7.15.2	**Q, B series with ELX-B (40%)**		
	Recharge BP and allow to stabilize. Reduce brake pipe to 448kPa (65psi). Confirm 5-6mm gap to the top of the bogie frame and valve arm and Empty indicator is protruding. Brake cylinder reading 159-179kPa (23-26psi) on the gauge		BP: Gap: BC:
	Recharge BP, install 20mm block under valve arm. Reduce BP pressure to 448kPa (65psi). Arm contacting block Valve empty indicator not protruding. Brake cylinder reading 400-427kPa (57-62psi) on the gauge		BP: Gap: BC:
7.15.3	**Wagons with EL60 valve (40%)**		
	Recharge brake pipe and allow to stabilize. Reduce the brake pipe to 448kPa (65psi). Confirm empty indicator has extended and Brake cylinder is 152-179kPa (22-26psi) on the gauge	Y / N	BP: BC:
	Recharge BP, install 20mm block under valve arm. Reduce BP pressure to 448kPa (65psi). Arm contacting block Valve empty indicator not protruding. Brake cylinder reading 400-427kPa (58-62psi) on the gauge	Y / N	BP: BC:
7.16	**End Cock Bleed test**		
	With wagon end cocks closed, apply air to BP at each end. Confirm air is blowing from bleed vents		
7.17	**Test Completion**		
	Make sure that the Brake Cylinder test gauge and other test devices have been removed from the car, including the test gauge fitted to the ELXB test port and Isolating cock on the J1 relay valve.		

Date		**Control Car Number**	
Maintainer Name		**SAP Number**	
This wagon has passed the brake test & is ready for traffic		**Signature**	

Internal Use Only
© 2023 Rio Tinto

Printed Copies are Uncontrolled

General Business / Affaires générales

Service Information & Sign-Off

Control Car #		**Completion Date:**	

North		**South**	
Confirm		**TECO**	
Supervisor Name:		**Signature:**	

Release to Traffic Checklist	**Responsible**	**Name**
Ensure Bearing Keepers & locking tabs are in Place and secure & tabs bent over	Last maintainer to Work on wagon	
Ensure retainer plates are in place & secure	Last maintainer to Work on wagon	
Ensure that hand brake operates freely & is oiled	Last maintainer to Work on wagon	
Check that yellow cable tie is fitted	Last maintainer to Work on wagon	
Ensure the removal of the RED TAG in the Rat trap if good for shunting	Last maintainer to Work on wagon	
Ensure TSPB valve is in the open position	Last maintainer to Work on wagon	
Walk round wagon for general once over inspection	Last maintainer to Work on wagon	
Ensure correct Wheel information is captured on the Data sheet	Supervisors	

RTIO-AM-0179836

Single Car Brake Test Sign-off sheet for ECP Overlay wagons

Amendment record

This table identifies the changes to the document made for each new version.

Document Control and revision history **Document type: Management Procedure**

RTIO-AM-0179836

Version	Date	Author	Reviewer	Approved by	Approved for use from
2.4.1	25 June 2015	Stephen Winter	Brooke Lane	Wayne Moss	03 June 2015
2.4.2	20 Dec 2016	Foha Rafiq	Ian Harding	John R. McArthur	06 September 2017
2.4.3	17 Oct 2017	Tatiana Joannides	Mason Cilia		
Issue 13	15 July 2020	Wynand Jordaan	Darren Brehmer	Declan Boyham	1 August 2020
Issue 14	4 Oct 2021	Wynand Jordaan	Mason Cilia	Declan Boyham	5 Oct 2021
Issue 15	1 Aug 2022	Wynand Jordaan	Matthew Wojas	Declan Boyham	18 Aug 2022
Issue 16	22 Feb 2022	Richard Bosworth, Wynand Jordaan, Ricco Dharmawan	Matthew Wojas	Declan Boyham	1 March 2023
Issue 17	09/05/2024	Wynand Jordaan	Andrew Stevens	Matthew Wojas	09/05/2024
Issue 18	07/01/2026	Wynand Jordaan	Shepherd Murape	Billie-Jo O'Brien	01/02/2026

Amendment Record **Document Area of Accountability: GM Railways Division**

Version & Date		Author	Change Management Number	Section Number	Modification Description
2.4.2	20 Dec 2016	Foha Rafiq	43621		Documents RTIO-AM-0105701and RTIO-AM-merged together. Reviewed Standard RTIO-LO-0001846 and included in Reference.
2.4.3	17 Oct 2017	Tatiana Joannides	43621		Brake Cylinder leakage test added as per RTIO-AM-0105701
Issue 13	15 July 2020	Wynand Jordaan			New BC Leak rates and general clean up
Issue 14	4 Oct 2021	Wynand Jordaan	81970		New EOTC reporting mark
Issue 15	1 Aug 2022	Wynand Jordaan			Update references and issue to line up with Standard
Issue 16	22 Feb 2023	Richard Bosworth, Wynand Jordaan, Ricco Dharmawan		Various	Pressure conversions updates and addition of Handheld tools electrical values A condition added when installing new lithium-ion battery. A note to re-do the ECP test when any components replaced is added. Wabtec battery load test section removed as it's irrelevant to the lithium-ion battery type.
Issue 17	09/05/2024	Wynand Jordaan	103365		Revision alignment to RTIO-LO-0001846
Issue 18	07/01/2026	Wynand Jordaan	131239	Various	Additional info on TSPB pre checks Brake Cylinder Leak test – additional requirement for RTCs Removal of using the ECP box for Hipot testing New TSPB reporting marks for ECP SA wagons – excluding EOTC's

1. Purpose and Scope

This document provides Sign-off on the Pneumatic Brake Test & Service for Ore Cars.

This document must be used in conjunction with the Single Car Brake test standard RTIO-LO-0001846.

This document will serve as a written record for the Brake test performed using the Rio Tinto Single Car Brake test standard – RTIO-LO-0001846

1. NOTE:

The revision of this Document must at all times match the revision of RTIO-LO-0001846, and can never be amended in isolation. RTIO-LO-0001846 controls the revision of this document.

- **Aim** — The aim of this task is to check, inspect, repair and maintain equipment for safety and reliability. These inspections are aimed at keeping the ore cars in operable condition and catching potential breakdowns before they become actual breakdowns.
- **Instructions** — Defects are to be recorded and relevant comments listed.
- **Isolations** — Isolations as per Ironsafe standards.
- **PPE Requirements** — PPE as per Ironsafe standards.
- **Tools & Equipment** — Tools and equipment to comply with Ironsafe standards.
- **Special Notes** — All inspectors shall tick all operations that they complete on the service sheet and sign off on the last page. Completed sheets are to be handed to the relevant site supervisor

Reference Documents	**FDMS No:**
1. Single Car Brake Test Standard	RTIO-LO-0001846

Reference	Test Equip.					
8.0		**ECP Database confirmation**	**Confirm Loaded**			
		Confirm that the correct ECP database has been loaded for ECP Overlay wagons. Car Database : 2019-Dec-03	2019_DEC_03			
8.1		**Cable & Fittings Inspections**	**Position**			
			1	**2**	**3**	**4**
		Inspect the inter car cables (ITC)				
		Inspect the BIW cable for damage				
		Inspect all male & female connectors for damage				
		Ensure that all cable clamps are in place & secure				
8.1		**CCD Junction Box & Warning Labels**	**Inspected**		**Repaired**	
		Inspect the condition of warning labels on ECP equipment.				
		Inspect CCID locking & securing bolts are secure.				
		Inspect the junction box & mounting plate is secure to body of wagon				
		Inspect the junction box lid bolts are secure and in place.				
8.1		**Armored Train Cable**	**Inspected**		**Repaired**	
		Inspect that the armored cable is secure to the body.				
		Inspect that the armored cable is not damaged.				
		Inspect armored train cable where it is entering CCID Box to ensure that it is not loose and seals correctly.				
		Inspect Train cable entering Junction Box to ensure it is not loose and seals correctly				
8.1		**ECP Manifold (Overlay and AEM)**	**Inspected**		**Repaired**	
		Inspect that the manifold cable is secure and not damaged.				
		Inspect the ECP manifold for damage.				
8.1		**Water ingress and battery PRT-RTIO-AM-0213664**				
		Ensure box clean and dry by removing excess dust/water				
		Spray inside of cabinet with approved electro contact cleaner				
		Confirm Battery protection foam pad are installed				
		Install new Lithium-Ion battery if fitted with old battery type				
		Confirm manifold cable has been injected with Neutral cure Silicon sealant to seal up potential water ingress				
		Confirm Velcro wire clips from the top of the door have been removed				
		Check that CCID box has foam gaskets on both the door and box. Ensure the gaskets are correctly adhered, free from damage				
		Ensure Breather valve has >5mm hole in the internal mesh to allow water flow.				
		Ensure that the box has a galvanized M12 washer that is 2 mm thick fitted to hinge				
		ECP Testing				
Note: The complete brake test must be re-done if any of the components are replaced during the inspection / testing process.						
8.3		**Train Line Continuity Test**				
		Resistance is between 5 and 8 ohms. Record the value	Pass / Fail		Ω	
8.4 & 8.5		**Hi Pot Tests**				
		Line to frame > 50 MΩ	Pass / Fail		Ω	
		Line to Line > 50 MΩ	Pass / Fail		Ω	
8.6		**Software Version Validation**	**Ver No:**		**Confirm**	
		Confirm Car Type is correct:				

Internal Use Only
© 2023 Rio Tinto

Printed Copies are Uncontrolled

General Business / Affaires générales

		Manifold version – 6.31	6.31	
		CCD version – 15.30	15.30	
		Car-Net version – 02.01	02.01	
		CID version – 02.20	02.20	
		Verify and record Manifold/CCD/CarNet/CID Software is the same as the versions indicated above, and upgrade if required		
		Confirm Car numbers in the configuration correspond with the actual wagon numbers.		
		Note: End Of Train Car's require the following format for its reporting mark - EOTC_/xxxxx e.g. EOTC_/15961 Where xxxxx is the Slave car number		
8.7.1		**ECP Minimum Application Test**		
		At 10% BC pressure is 48 ± 14kPa (7 ± 2psi)		
8.7.2		**ECP Full-Service Application Test**		
		At 100% BC pressure is 441 ± 14kPa (64 ± 2psi)		
8.7.3		**ECP Emergency Test**		
		At 120% BC pressure is 517 ± 7kPa (75 ± 1psi)		
8.7.4		**Brake Cylinder Leakage Test**		
		Initial brake pressure		
		Brake pressure after 1 minute		
		Allowance BC pressure change +/- 1 psi or 7 kPa		
8.7.6		**Manual Release Test**		
		Verify brake pistons retract		
		Verify CCD Cut-out		
		Verify CCD Cut-in		
8.7.8		**Battery Operations Test**		
		Brake cylinder pressure is 64 +/- 2 psi (441kpa +/-14kpa)		
		Car remains operable		
		Battery is above 12V		
		Verify both LED's flashing		
		Brake cylinder pressure decreases to 0psi/kpa		
		Record the Battery Installation date		/ /
		Record the Battery Last Charged date		/ /
		Inspect that the battery cables are secure and not corroded		
		Inspect that the battery is secure and not leaking		
8.7.9		**Wabtec CCD Hardware Test**		**Pass** / **Fail**
	C/N 920	Pass voltage is > 12.5 VDC, Fail voltage is <= 12.5 VDC The voltage on the meter should be between 12.9 and 13.1 VDC. If the voltage is <= 12.5 VDC, then replace CCD.		
	C/N 920	Pass voltage is > 12.5 VDC, Fail voltage is <= 12.5V DC The voltage on the meter should be between 13.8 and 14.0 VDC If the voltage is <= 12.5 VDC, then replace CCD.		
8.7.10		**ECP Shutdown and Manual Release**		
		No exceptions report		
		Verify LED's off		
		Verify pistons retract		

Date		**Control Car Number**	
Maintainer Name		**SAP Number**	
This wagon has passed the brake test & is ready for traffic		**Signature**	

Internal Use Only
© 2023 Rio Tinto

Printed Copies are Uncontrolled

General Business / Affaires générales